



南开大学  
Nankai University



天津市智能机器人技术重点实验室  
Tianjin Key Laboratory of Intelligent Robotics  
南开大学机器人与信息自动化研究所  
Institute of Robotics & Automatic Information System

# 第六章 机器人动力学

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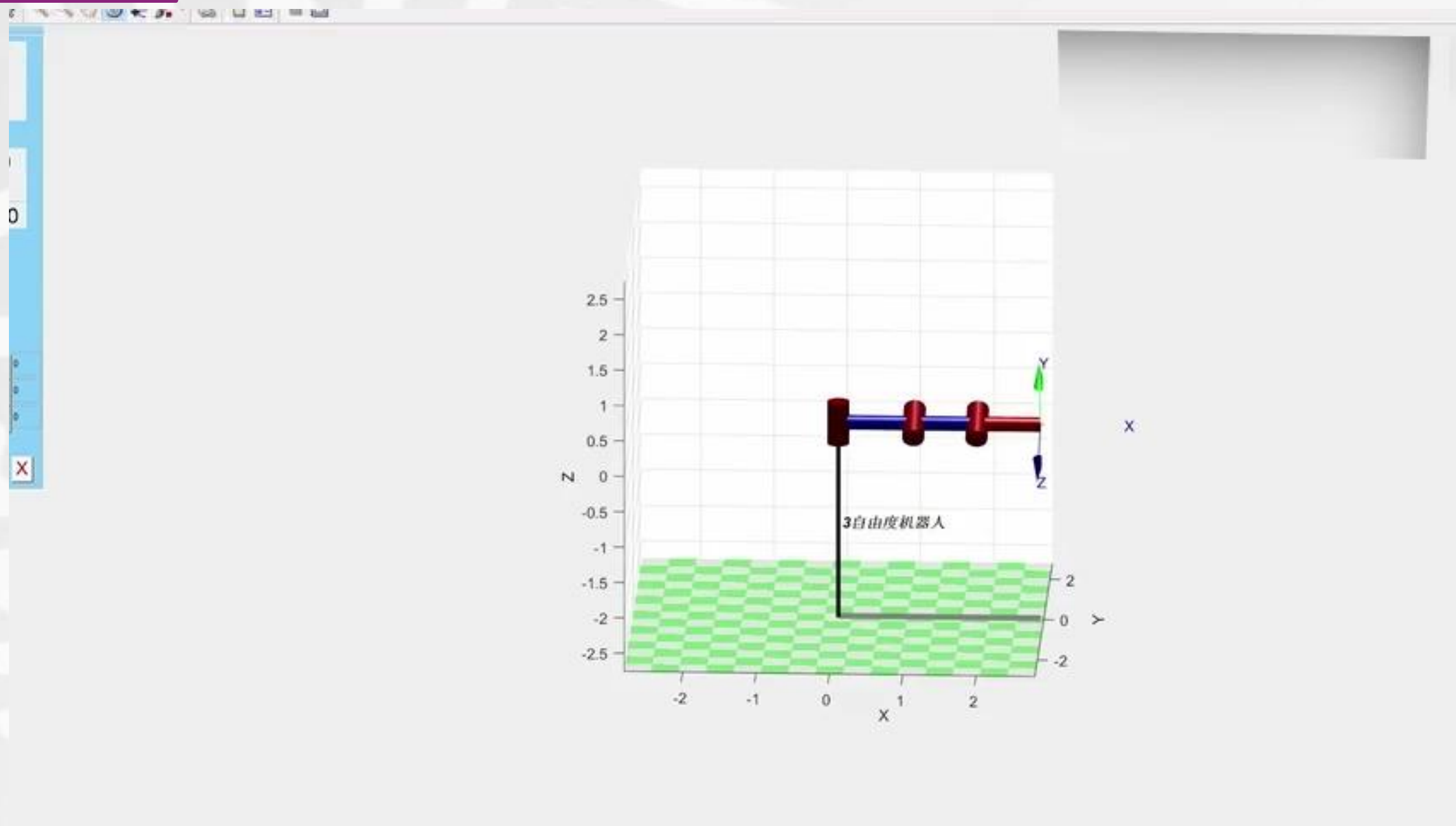
## 《机器人学导论》

袁明星 副教授

Email: [mxyuan@nankai.edu.cn](mailto:mxyuan@nankai.edu.cn)

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# 动画与仿真



# 课程内容



## 1. 刚体动力学



## 2. 牛顿-欧拉动力学方程



## 3. 拉格朗日动力学算法

# 1.1 机器人动力学概念

## ■ 基本概念

- 机器人动力学是多刚体紧耦合动态系统——模型十分复杂

## ■ 为具有刚性臂机器人建立数学模型的两种理论：

- 牛顿-欧拉方程（受力分析）

- 拉格朗日力学，特别是二阶拉格朗日方程（基于能量的动力学方法）

## ■ 基本应用

- For computer simulation

- Design of suitable controller

- Evaluation of robot structure



# 1.2 运动物体受力分析

- 空间中一物体要改变其运动状态(即产生加速度):

$$F_1 = ma$$

- 在地球上受重力作用:  $G$

- 如果还有转动:

- 离心力:

- 旋转参考系下 (非惯性参考系) 观测到的一种惯性力, **假想的力**, 方向: 沿半径背离圆心

$$F_2 = m\omega^2 r$$

- 科里奥利力:

- 旋转参考系下 (非惯性参考系), 物体相对于参考系运动(相对速度为 $v'$ )观测到的一种惯性力, **假想的力**

$$F = 2mv' \times \omega$$

- 最终, 物体受到的合力为:

$$\Gamma = ma + G + m\omega^2 r + 2mv' \times \omega$$

**假想的力:** 非惯性系中, 利用牛顿力学进行分析时引入的, 并非真实存在的力

# 1.3 科里奥利力(Coriolis force)

## Labortory Demonstrations of Planetary-Style Fluid Dynamics

Jonathan Aurnou & John Cantwell  
UCLA Earth & Space Sciences

*spinlabucla*  
SIMULATED PLANETARY INTERIOR LAB

*spinlabucla*

# 1.4 机器人动力学基本形式

- 机器人动力学方程：

$$M(q)\ddot{q} + V(q, \dot{q}) + G(q) = \Gamma$$

- $q$ ：机器人关节空间坐标,  $q = [q_1, q_2, \dots, q_n]^T$
- $\dot{q}, \ddot{q}$ ：机器人关节速度与加速度
- $\Gamma$ ：广义关节力矩矢量
- $M(q)$ ：机器人惯性阵
- $V(q, \dot{q})$ ：离心力和科里奥利力
- $G(q)$ ：重力

# 1.5 质点直线运动

- 质点：有质量无大小
- 牛顿第二定律：

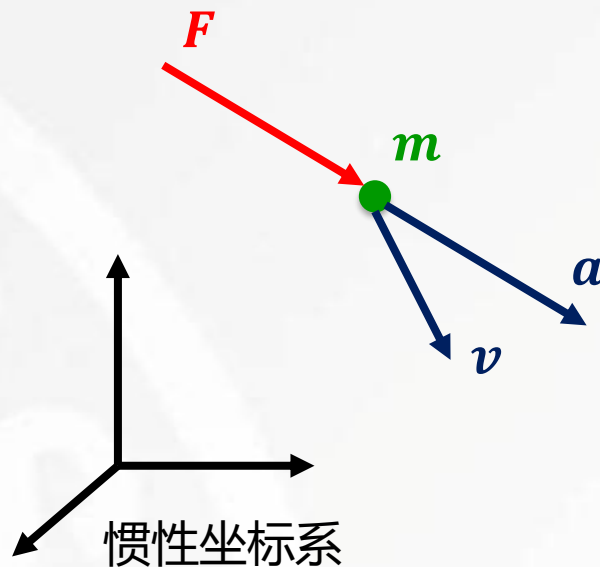
$$F = ma \xrightarrow{a=\dot{v}} F = \frac{d}{dt}(mv)$$

- 动量：

$$Q = mv$$

- 动量守恒定律：一个系统不受外力或所受外力之和为零，这个系统的总动量保持不变。
- 动量的变化率等于所受外力

$$F = \dot{Q}$$



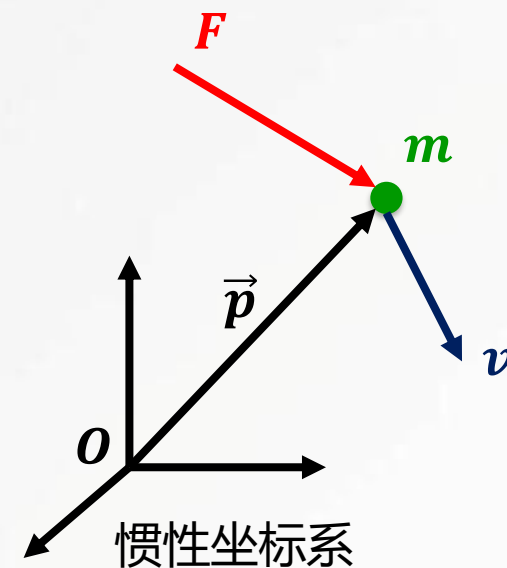


# 1.6 质点旋转运动

- 质点：有质量无大小
- 角动量  $L = \vec{p} \times m\vec{v}$
- 角动量守恒定律：一个系统不受外力或所受外力之和为零或合外力指向参考系的原点，这个系统的总角动量保持不变。

$$\begin{aligned}\frac{dL}{dt} &= m\dot{\vec{p}} \times \vec{v} + m\vec{p} \times \dot{\vec{v}} = m\vec{v} \times \vec{v} + m\vec{p} \times \dot{\vec{v}} = \vec{p} \times (m\dot{\vec{v}}) \\ &= \vec{p} \times \vec{F} = \tau\end{aligned}$$

- 角动量的变化率等于所受外力矩:  $\dot{L} = \tau$



# 1.7 刚体直线运动

## ■ 刚体直线运动:

- 空间平移不变性: 刚体上所有点的线速度相同

$$v_i = v$$

## ■ 质点线动量 $Q_i$

$$Q_i = m_i v_i$$

## ■ 整个刚体的线动量 $Q$ :

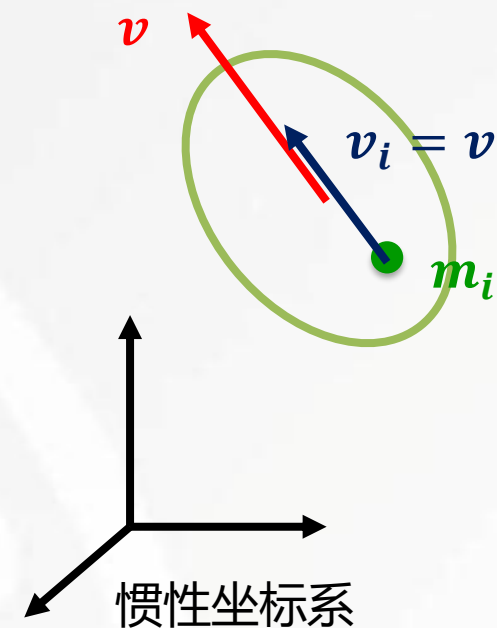
$$Q = \sum_i Q_i = \sum_i m_i v_i$$

$$\xrightarrow{m_i = \rho dV} Q = \int_V v_i \rho dV = \left( \int_V \rho dV \right) v$$

刚体的质量 $m$

- 刚体的质量 $m$ : 刚体空间运动的空间各向同性

$$m = \begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & m \end{bmatrix}$$



$$Q = mv$$

# 1.8 刚体旋转运动

## ■ 刚体旋转运动:

□ 空间旋转不变性: 刚体上所有点的角速度相同:

$$v_i = \omega \times p_i$$

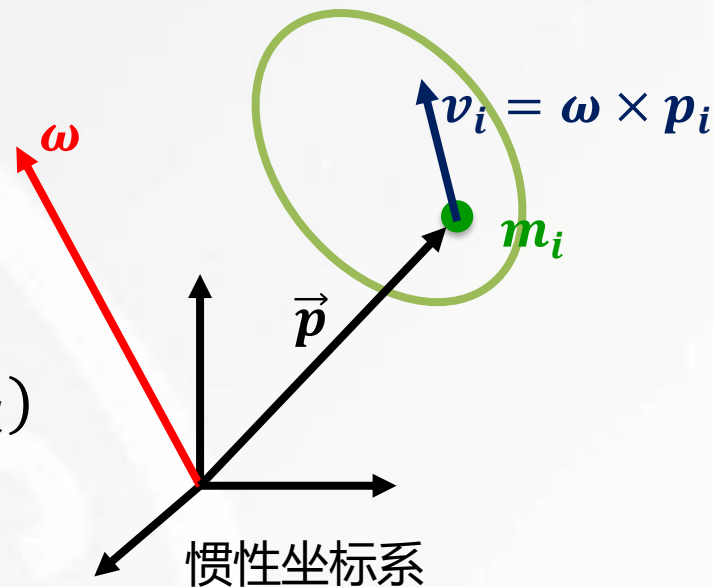
## ■ 质点角动量

$$L_i = \vec{p}_i \times m_i v_i = m_i p_i \times (\omega \times p_i)$$

## ■ 整个刚体的角动量:

$$L = \sum_i L_i = \sum_i m_i p_i \times (\omega \times p_i)$$

$$\xrightarrow{m_i = \rho dV} L = \int_V p \times (\omega \times p) \rho dV = \int_V -p \times (p \times \omega) \rho dV = \left( \int_V -\hat{p} \hat{p} \rho dV \right) \omega$$

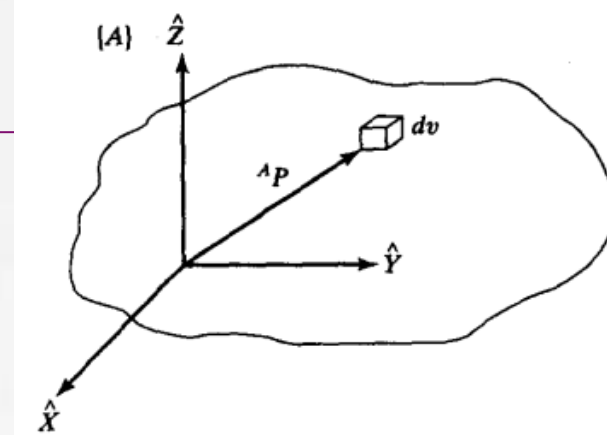


刚体的惯性张量  $I$

$$L = I\omega$$

# 1.9 刚体惯性张量

■ 刚体惯性张量  $I$ :



$$I = \int_V -\hat{p}\hat{p} \rho dV$$

$$p = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \rightarrow \hat{p} = \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix} \Rightarrow -\hat{p}\hat{p} = \begin{bmatrix} y^2 + z^2 & -xy & -xz \\ -xy & z^2 + x^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{bmatrix}$$

惯量矩(转动惯量)

惯量积

惯量张量

$$I = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{bmatrix}$$

$$I_{xx} = \iiint (y^2 + z^2) \rho dx dy dz$$

$$I_{yy} = \iiint (z^2 + x^2) \rho dx dy dz$$

$$I_{zz} = \iiint (x^2 + y^2) \rho dx dy dz$$

$$I_{xy} = \iiint (xy) \rho dx dy dz$$

$$I_{yz} = \iiint (yz) \rho dx dy dz$$

$$I_{xz} = \iiint (xz) \rho dx dy dz$$

# 1.9 刚体惯性张量

$$p = [x \quad y \quad z]^T$$

$$p^T p = x^2 + y^2 + z^2 \quad pp^T = \begin{bmatrix} x \\ y \\ z \end{bmatrix} [x \quad y \quad z] = \begin{bmatrix} x^2 & xy & xz \\ xy & y^2 & yz \\ xz & yz & z^2 \end{bmatrix}$$

$$-\hat{p}\hat{p} = (p^T p)I_3 - pp^T$$

惯量矩: 刚体绕3个主轴的转动惯量      惯量积:

$$I_{xx} = \iiint (y^2 + z^2) \rho dx dy dz$$

$$I_{yy} = \iiint (z^2 + x^2) \rho dx dy dz$$

$$I_{zz} = \iiint (x^2 + y^2) \rho dx dy dz$$

$$I_{xy} = \iiint (xy) \rho dx dy dz$$

$$I_{yz} = \iiint (yz) \rho dx dy dz$$

$$I_{xz} = \iiint (xz) \rho dx dy dz$$

惯量矩和惯量积与选定的参考坐标系有关，当选定的坐标系使得惯量积为零时，该参考坐标系的轴称为**主轴**，相应的惯量矩称为**主惯量矩**。

# 1.9 刚体惯性张量的一般理解

假设有一个质点围绕  $x$  轴以角速度  $\omega_x$  旋转，我们来求该质点分别对  $x$  轴， $y$  轴， $z$  轴的动量矩。

假设由坐标系原点指向质点的向量为  $r = [x, y, z]^T$

质点的速度为：  $v = \omega_x [1, 0, 0]^T \times r = [0, -\omega_x z, \omega_x y]^T$

假设质点的质量为  $m$ ，质点相对于原点的动量矩为：

$$L = r \times mv = [m\omega_x(z^2 + y^2), m\omega_x(-xy), m\omega_x(-xz)]^T$$

所以，质点对于  $x$  轴的动量矩为：  $m\omega_x(z^2 + y^2)$

对于  $y$  轴的动量矩为：  $m\omega_x(-xy)$

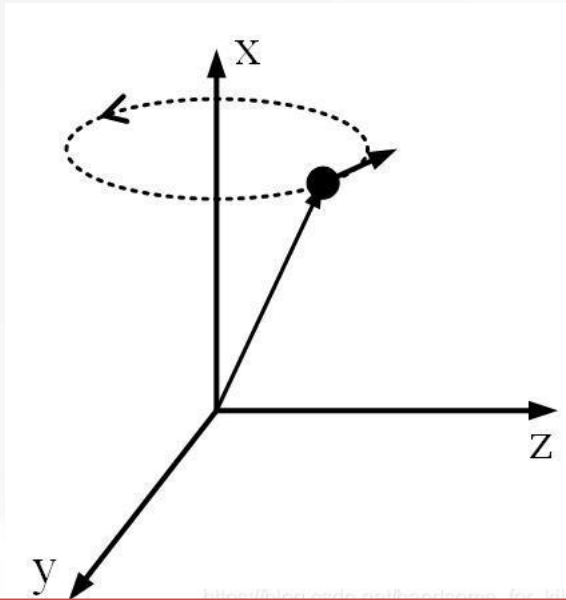
对于  $z$  轴的动量矩为：  $m\omega_x(-xz)$

动量矩为：  $L = I\omega$

所以，质点的相对于  $x$  轴的惯性矩为：  $m(z^2 + y^2)$

对于  $y$  轴的惯性积为：  $m(-xy)$

对于  $z$  轴的惯性积为：  $m(-xz)$



惯性张量的第一列就代表刚体沿  $x$  轴旋转时，对  $x$  轴， $y$  轴， $z$  轴产生的动量矩的大小。

# 1.9 刚体惯性张量相关定理

## 平行轴定理

设刚体质量为 $m$ ，绕通过质心转轴的转动惯量为 $I_c$ ，将此轴朝任何方向平行移动一个距离 $p_c$ ，则绕新轴的转动惯量 $I_A$ 为：

$$I_A = I_c + m[(p_c^T p_c)I_3 - p_c p_c^T]$$

$$I_{Axx} = I_{Cxx} + m(y_c^2 + z_c^2),$$

$$I_{Axy} = I_{Cxy} + mx_c y_c,$$

$$I_{Ayy} = I_{Cyy} + m(x_c^2 + z_c^2),$$

$$I_{Ayz} = I_{Cyz} + my_c z_c,$$

$$I_{Azz} = I_{Czz} + m(x_c^2 + y_c^2),$$

$$I_{Axz} = I_{Cxz} + mx_c z_c$$

$p_c = [x_c, y_c, z_c]^T$ 为质心在平移后新坐标系下的位置矢量， $x_c, y_c, z_c$ 即为质心在平移后新坐标系下的坐标分量。



# 1.9 刚体惯性张量相关定理

## 垂直轴定理

一个平面刚体薄片对于垂直它的平面的轴的转动惯量，等于绕平面内与垂直轴相交的任意两正交轴的转动惯量之和。

$$I_Z = I_X + I_Y$$

假设刚体是一块薄片（厚度可以忽略不计），密度均匀。设定薄片的面与  $X - Y$  共平面；那么，刚体对于  $X$  轴、 $Y$  轴、 $Z$  轴的转动惯量分别为

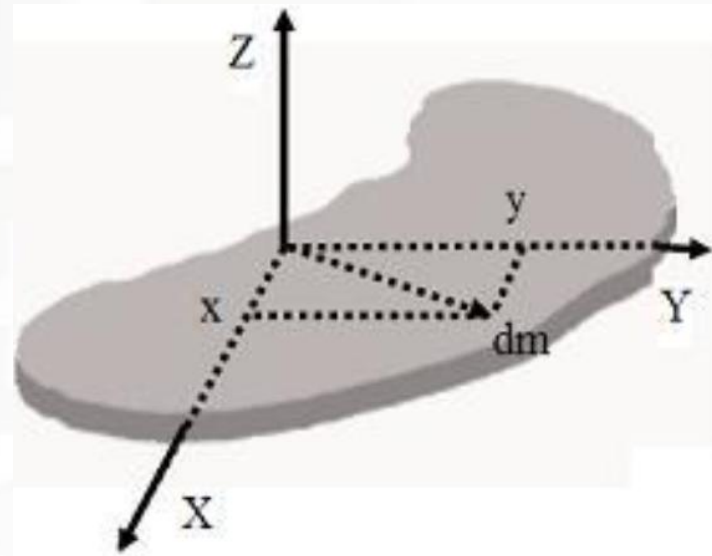
$$I_X = \int (z^2 + y^2) dm \quad I_Y = \int (x^2 + z^2) dm \quad I_Z = \int (x^2 + y^2) dm$$

忽略  $z$ ，有

$$I_X \approx \int y^2 dm \quad I_Y \approx \int x^2 dm$$

所以

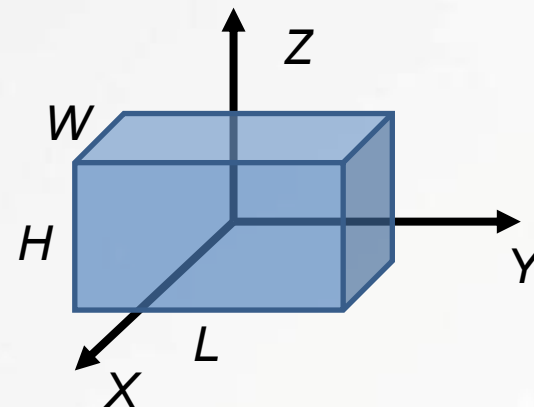
$$I_Z = I_X + I_Y$$





# 1.9 刚体惯性张量求解(1)

一均匀密度长方体，其长宽高依次为 $L, W, H$ ，其密度为 $\rho$ ，坐标系 $OXYZ$ 原点和质心重合，其 $X$ 轴,  $Y$ 轴,  $Z$ 轴分别平行于长方体的宽、长和高，求此长方体在坐标系 $OXYZ$ 下的惯性张量惯量矩：



$$I_{xx} = \int_{-\frac{H}{2}}^{\frac{H}{2}} \int_{-\frac{L}{2}}^{\frac{L}{2}} \int_{-\frac{W}{2}}^{\frac{W}{2}} (y^2 + z^2) \rho dx dy dz = \frac{H^2 + L^2}{12} LWH\rho = \frac{m}{12} (H^2 + L^2)$$

$$I_{yy} = \frac{m}{12} (H^2 + W^2), I_{zz} = \frac{m}{12} (W^2 + L^2)$$

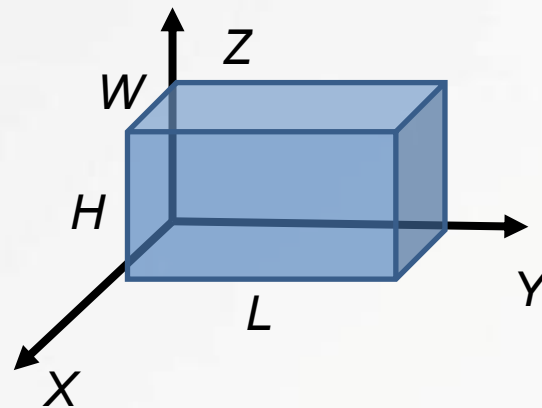
$$I = \begin{bmatrix} \frac{m}{12} (H^2 + L^2) & 0 & 0 \\ 0 & \frac{m}{12} (H^2 + W^2) & 0 \\ 0 & 0 & \frac{m}{12} (W^2 + L^2) \end{bmatrix}$$

惯量积：

$$I_{xy} = \int_{-\frac{H}{2}}^{\frac{H}{2}} \int_{-\frac{L}{2}}^{\frac{L}{2}} \int_{-\frac{W}{2}}^{\frac{W}{2}} (xy) \rho dx dy dz = 0, I_{yz} = 0, I_{xz} = 0$$

## 1.9 刚体惯性张量求解(2)

一均匀密度长方体，长宽高为 $L, W, H$ ，其密度为 $\rho$ ，坐标系 $OXYZ$ 原点和长方体顶点重合，其 $X$ 轴， $Y$ 轴， $Z$ 轴分别平行于长方体的宽、长和高，求此长方体在坐标系 $OXYZ$ 下的惯性张量



平行轴原理:  $I_A = I_C + m[(p_C^T p_C)I_3 - p_C p_C^T]$

$$I_{Axx} = I_{Cxx} + m(y_c^2 + z_c^2), \quad I_{Ayy} = I_{Cyy} + m(x_c^2 + z_c^2), \quad I_{Azz} = I_{Czz} + m(x_c^2 + y_c^2),$$

$$I_{Axy} = I_{Cxy} + mx_c y_c, \quad I_{Ayz} = I_{Cyz} + my_c z_c, \quad I_{Axz} = I_{Cxz} + mx_c z_c$$

$$x_c = \frac{W}{2}, \quad y_c = \frac{L}{2}, \quad z_c = \frac{H}{2}$$

$$I = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{bmatrix}$$

$$I_{Axx} = I_{Cxx} + m(y_c^2 + z_c^2) = \frac{m}{12}(H^2 + L^2) + \frac{m}{4}(L^2 + H^2) = \frac{m}{3}(H^2 + L^2)$$

$$I_{Ayy} = I_{Cyy} + m(x_c^2 + z_c^2) = \frac{m}{3}(H^2 + W^2), \quad I_{Azz} = I_{Czz} + m(x_c^2 + y_c^2) = \frac{m}{3}(W^2 + L^2)$$

$$I_{Axy} = I_{Cxy} + mx_c y_c = \frac{m}{4}WL, \quad I_{Ayz} = I_{Cyz} + my_c z_c = \frac{m}{4}LH, \quad I_{Axz} = I_{Cxz} + mx_c z_c = \frac{m}{4}WH$$

# 1.9 刚体惯性张量性质

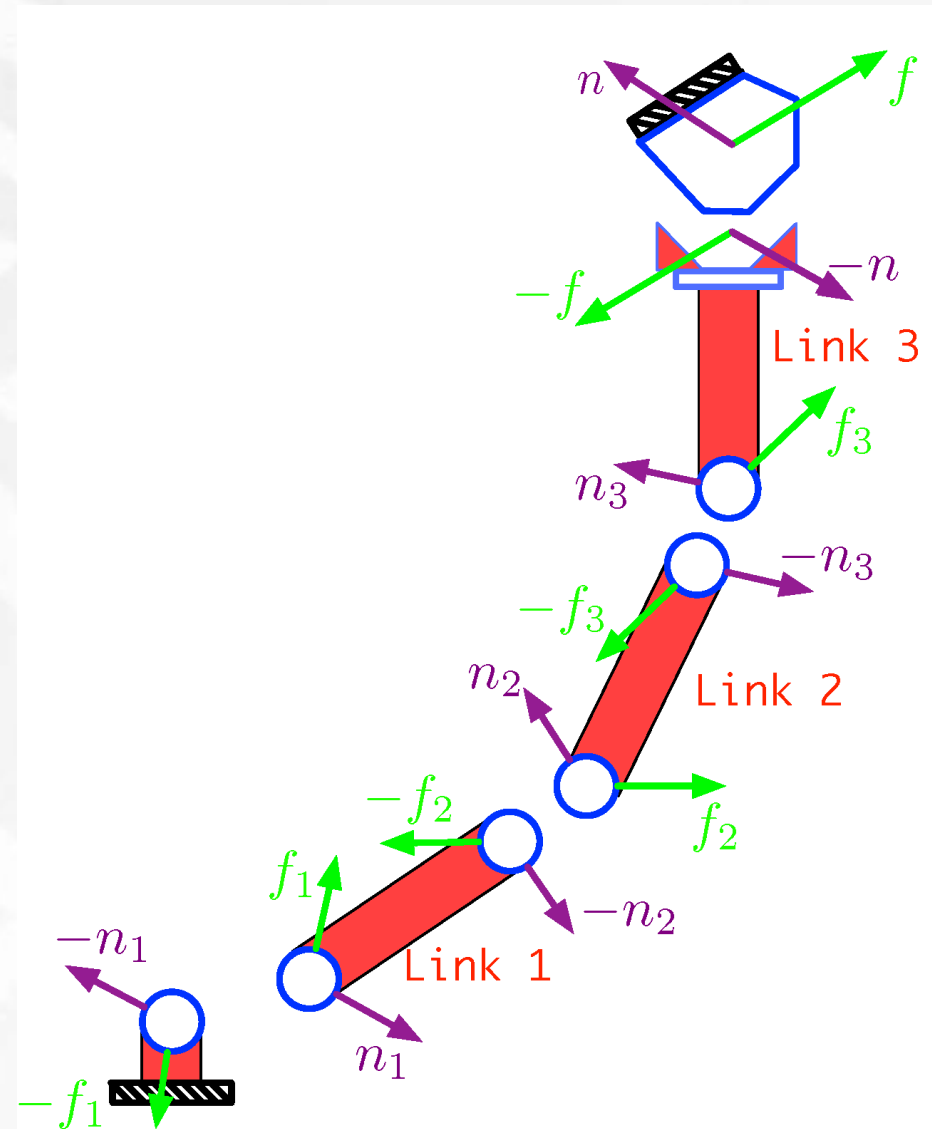
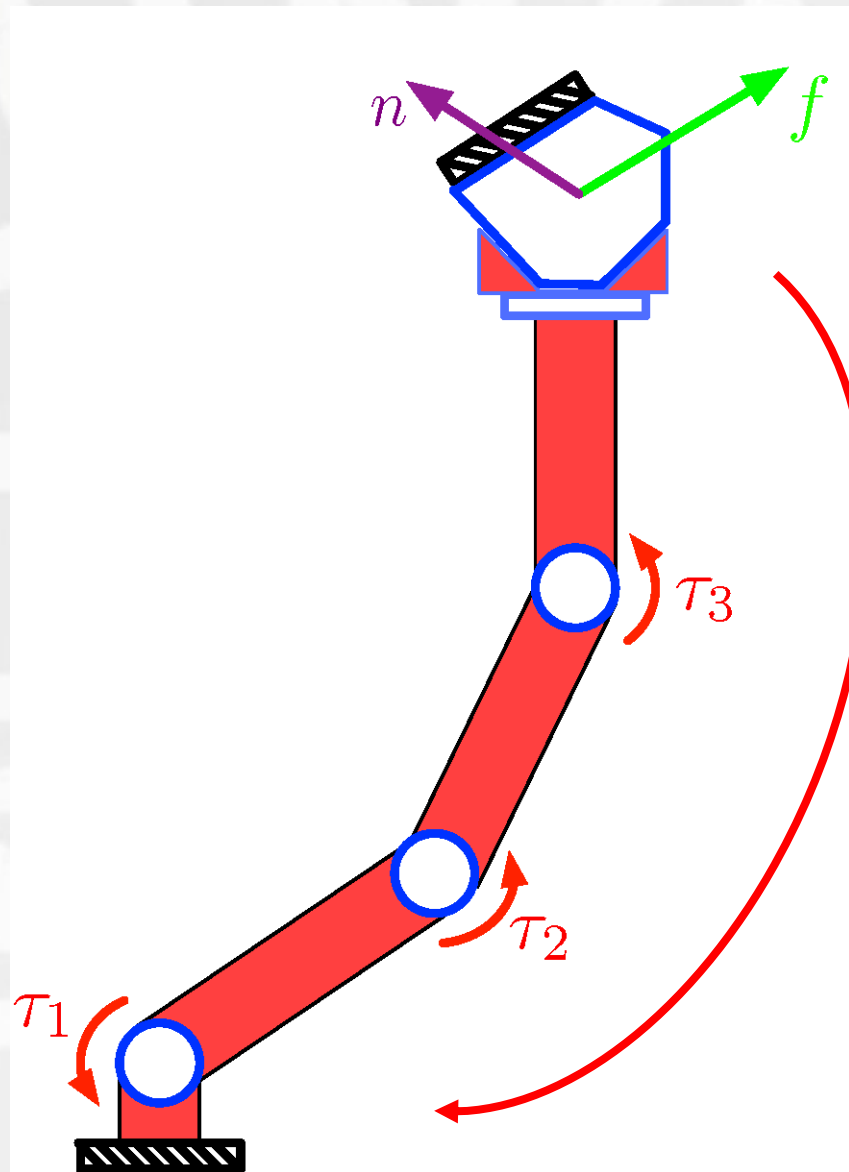
1. 如果坐标系的两个坐标轴构成的平面为刚体质量分布的对称平面，则垂直于这个对称平面的坐标轴与另一个坐标轴的惯性积为0
2. 惯性矩永远为正值。惯性积可能为正值也可能为负值
3. 无论参考坐标系姿态如何变化，三个惯性矩的和保持不变——惯性张量的迹不变
4. 惯性张量的特征值为刚体的主惯性矩，相应的特征矢量为主轴
5. 转动惯量只决定于刚体的形状、质量分布和**转轴的位置**，而同刚体绕轴的转动状态（如角速度的大小）无关
6. 在一个空间直角坐标系中，若物体的三个惯性积有两个为零，则其公用坐标轴，称为“惯性主轴”
7. 惯性主轴是物体“特定转轴”的固有性质，**与坐标系的选择无关**。
8. **刚体对质心的转动惯量是最小**

02

# 牛顿-欧拉动力学方程

Newton-Euler Dynamics Algorithm

## 2.1 机器人受力分析



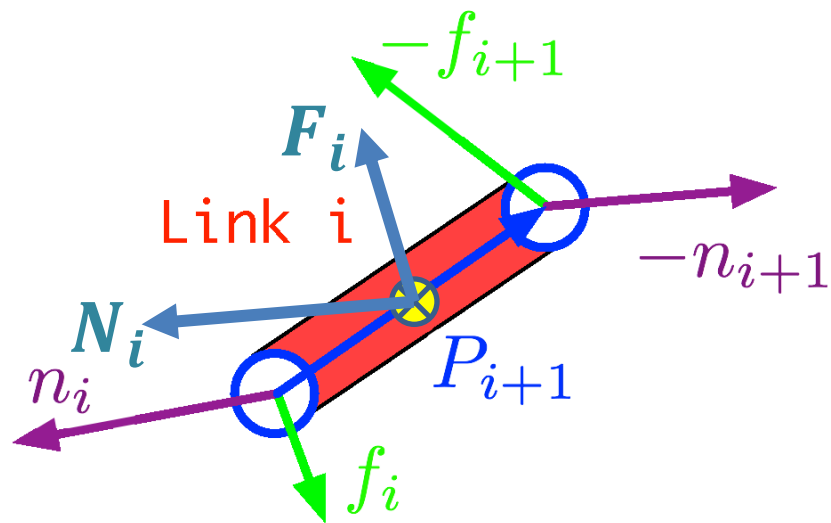
## 2.1 机器人受力分析

迭代求解:

$$f_i = f_{i+1} + F_i$$

$$n_i = n_{i+1} + P_{i+1} \times f_{i+1} + P_{ci} \times F_i + N_i$$

$$\tau_i = \begin{cases} n_i^T \cdot Z_i & \text{旋转关节} \\ f_i^T \cdot Z_i & \text{平移关节} \end{cases}$$



## 2.2 牛顿-欧拉动力学方程

### 牛顿方程

$$\frac{d(m\vec{v})}{dt} = \vec{f} \xrightarrow{\text{定常质量的刚体}} m\vec{a} = \vec{f}$$

$m$  is the mass of the body,  $v$  is the velocity of the center of mass with respect to an inertial frame, and  $f$  is the sum of external forces applied to the body.

$\vec{a} = \dot{\vec{v}}$  is the acceleration of the center of mass.



## 2.2 牛顿-欧拉动力学方程

欧拉方程

$$\frac{d(I_0 \vec{\omega}_0)}{dt} = \vec{\tau}_0$$

$I_0$  is the moment of inertia of the body about an **inertial frame** whose origin is at the center of mass,  $\omega_0$  is the angular velocity of the body, and  $\tau_0$  is the sum of torques applied to the body

Suppose we attach a frame rigidly to the body, and let  $I$  denote the inertia matrix of the body with respect to this frame. Let  $R$  denote the orientation of the frame rigidly attached to the body with respect to the inertial frame, then

$$I_0 = RIR^T$$

$$\text{证明: } I_0 \omega_0 = L_0 = RL = R(I\omega) = RI(R^T \omega_0)$$

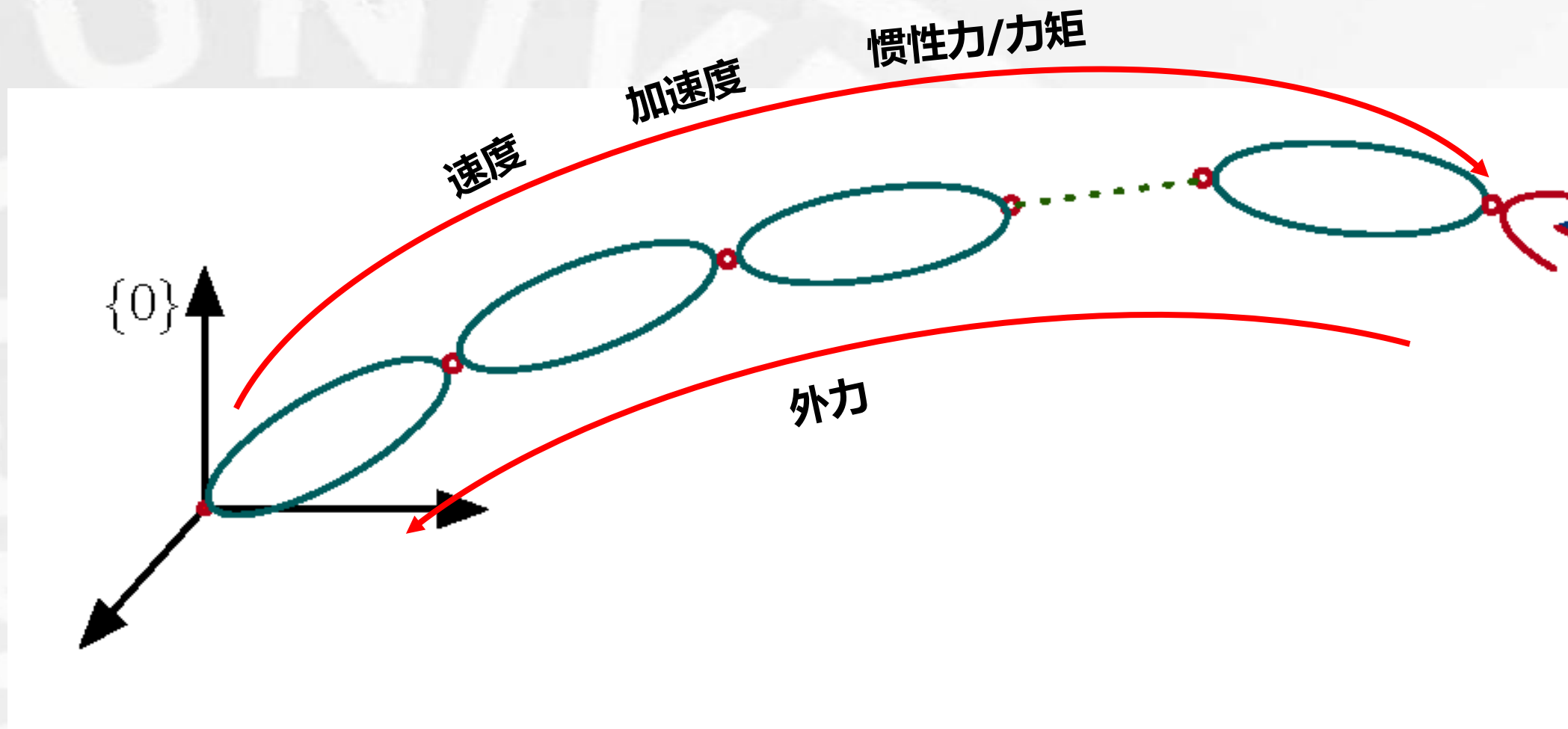
## 2.2 牛顿-欧拉动力学方程

$$\omega_0 = R\omega \implies I_0\omega_0 = I_0R\omega \xrightarrow{I_0 = RIR^T} I_0\omega_0 = RIR^T R\omega = RI\omega$$

$$\begin{aligned} R\tau = \tau_0 &= \frac{d(I_0\omega_0)}{dt} \xrightarrow{I_0\omega_0 = RI\omega} R\tau = \frac{d(RI\omega)}{dt} = \overset{\dot{R} = S(\omega_0)R}{\dot{R}I\omega} + RI\dot{\omega} = \overset{\omega_0 = R\omega}{S(\omega_0)RI\omega} + RI\dot{\omega} \\ &= \overset{S(R\omega) = RS(\omega)R^T}{S(R\omega)RI\omega} + RI\dot{\omega} = RS(\omega)R^T RI\omega + RI\dot{\omega} \\ &= RS(\omega)I\omega + RI\dot{\omega} = R[(\omega \times I\omega) + I\dot{\omega}] \end{aligned}$$

$$\tau = (\omega \times I\omega) + I\dot{\omega}$$

## 2.3 机器人动力学建模过程



## 2.3.1 连杆速度与加速度

坐标系与符号约定：

$\{0\}$ ：惯性基坐标系（固定坐标系）

$\{i\}$ ：第 $i$ 连杆连体坐标系

左上标： $^A x$  表示矢量在坐标系  $A$  下表示

${}^0\omega_i$ ：连杆  $i$  相对惯性系  $\{0\}$  的角速度矢量（物理矢量）

${}^i\omega_i$ ： ${}^0\omega_i$  在连体系  $\{i\}$  下的表达

绝对导数  $\frac{d}{dt}(\cdot)$ ：在惯性系  $\{0\}$  下求导

相对导数  $\frac{d^*}{dt}(\cdot)$ ：在当前连体坐标系下直接求导

$R_{i,i+1}:\{i+1\} \rightarrow \{i\}$  旋转矩阵

## 2.3.1 连杆速度与加速度

对动系{k}下描述的矢量  ${}^k\mathbf{u}$

$$\frac{d}{dt}{}^k\mathbf{u} = \frac{d^*}{dt}{}^k\mathbf{u} + {}^k\boldsymbol{\omega}_k \times {}^k\mathbf{u}$$

$\frac{d}{dt}{}^k\mathbf{u}$  : 矢量物理绝对导数, 在动系{i}下描述;

$\frac{d^*}{dt}{}^k\mathbf{u}$  : 动系内相对导数 (观察者随动系转, 直接对分量求导)

${}^{i+1}\mathbf{e}_z = [0 \quad 0 \quad 1]^T$ , 在{i+1}为常矢量。

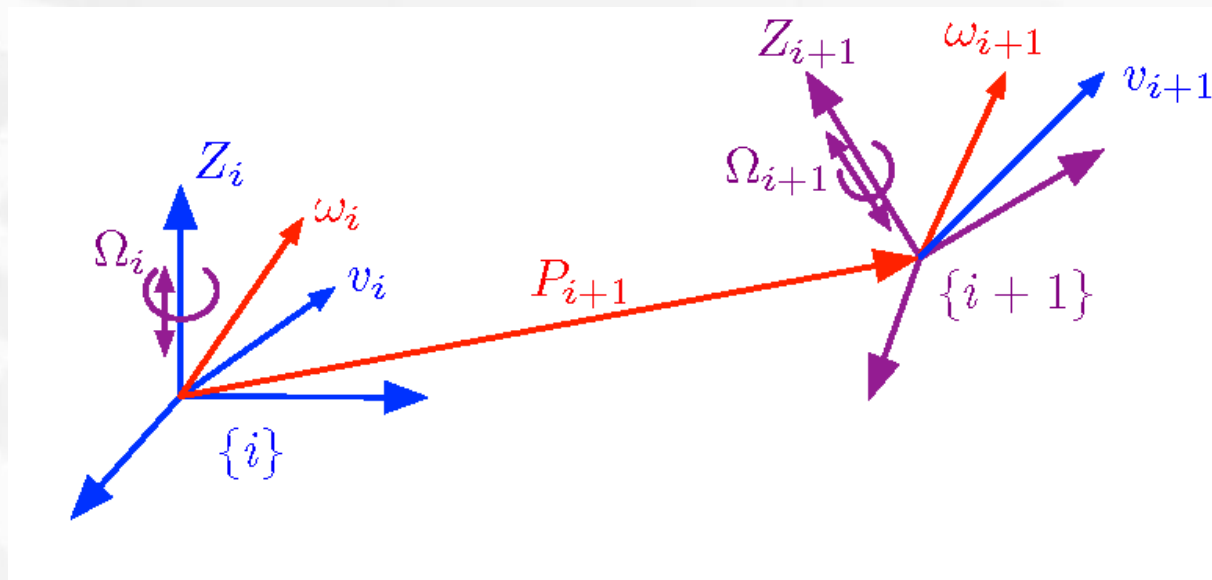
## 2.3.1 连杆速度与加速度

角速度迭代:

$$\omega_{i+1} = \omega_i + \Omega_{i+1} = \omega_i + \dot{\theta}_{i+1} \cdot Z_{i+1}$$

角加速度计算: (惯性系下定义)

$$\begin{aligned} {}^{i+1}\dot{\omega}_{i+1} &= {}^{i+1}\dot{\omega}_i + \frac{d}{dt}({}^{i+1}\dot{\theta}_{i+1} Z_{i+1}) \\ &= {}^{i+1}\dot{\omega}_i + \ddot{\theta}_{i+1} \cdot {}^{i+1}Z_{i+1} + \dot{\theta}_{i+1} \cdot {}^{i+1}\omega_i \times {}^{i+1}Z_{i+1} \end{aligned}$$



## 2.3.1 连杆速度与加速度

速度迭代:

$$v_{i+1} = v_i + \omega_i \times P_{i+1} + \dot{d}_{i+1} \cdot Z_{i+1}$$

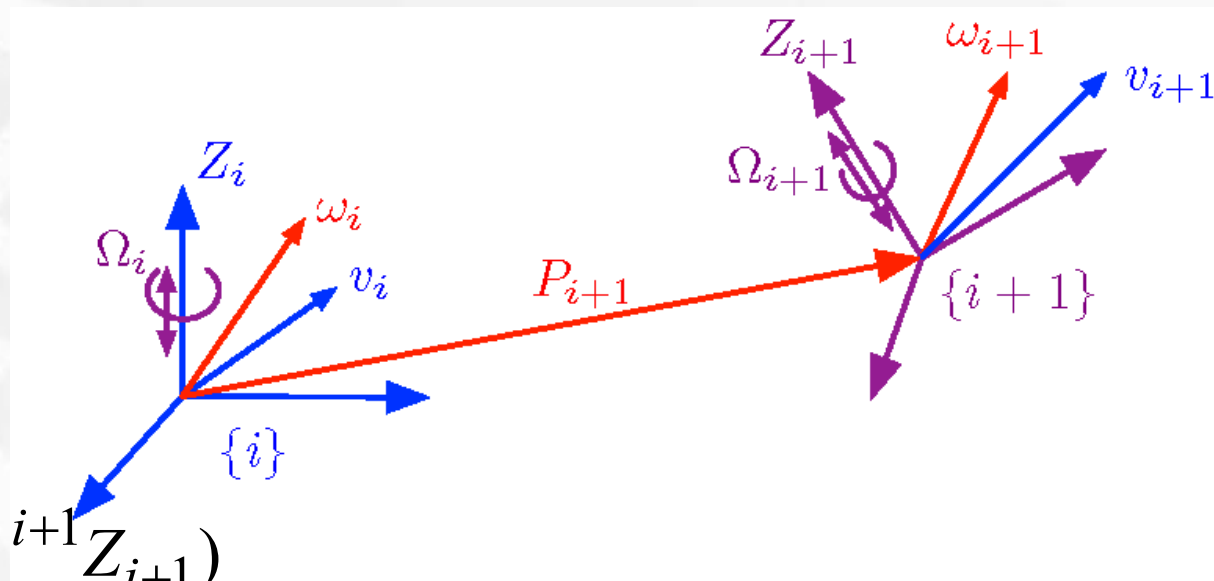
加速度计算: (惯性系下定义)

$${}^{i+1}\dot{v}_{i+1} = {}^{i+1}\dot{v}_i + \frac{d}{dt}({}^{i+1}\omega_i \times {}^{i+1}P_{i+1}) + \frac{d}{dt}(\dot{d}_{i+1} \cdot {}^{i+1}Z_{i+1})$$

$$\frac{d}{dt} {}^{i+1}\omega_i \times {}^{i+1}P_{i+1} + {}^{i+1}\omega_i \times \frac{d}{dt} {}^{i+1}P_{i+1}$$

$$= {}^{i+1}\dot{\omega}_i \times {}^{i+1}P_{i+1} + {}^{i+1}\omega_i \times (\dot{d}_{i+1} \cdot {}^{i+1}Z_{i+1} + {}^{i+1}\omega_i \times {}^{i+1}P_{i+1})$$

$${}^{i+1}\dot{v}_{i+1} = {}^{i+1}\dot{v}_i + {}^{i+1}\dot{\omega}_i \times {}^{i+1}P_{i+1} + \underbrace{{}^{i+1}\omega_i \times ({}^{i+1}\omega_i \times {}^{i+1}P_{i+1})}_{\text{离心加速度}} + \underbrace{2\dot{d}_{i+1} \cdot {}^{i+1}\omega_i \times {}^{i+1}Z_{i+1}}_{\text{科里奥利加速度}} + \ddot{d}_{i+1} \cdot {}^{i+1}Z_{i+1}$$



$$\ddot{d}_{i+1} \cdot {}^{i+1}Z_{i+1} + \dot{d}_{i+1} \cdot {}^{i+1}\omega_i \times {}^{i+1}Z_{i+1}$$

离心加速度

科里奥利加速度



## 2.3.2 连杆惯性力计算

连杆质心线速度：

$$v_{C_{i+1}} = v_{i+1} + \omega_{i+1} \times P_{C_{i+1}}$$

连杆质心线加速度：

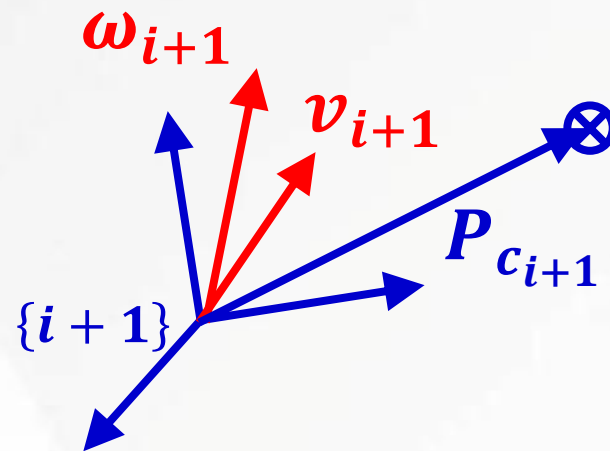
$$\dot{v}_{C_{i+1}} = \dot{v}_{i+1} + \dot{\omega}_{i+1} \times P_{C_{i+1}} + \omega_{i+1} \times (\omega_{i+1} \times P_{C_{i+1}})$$

连杆惯性力：

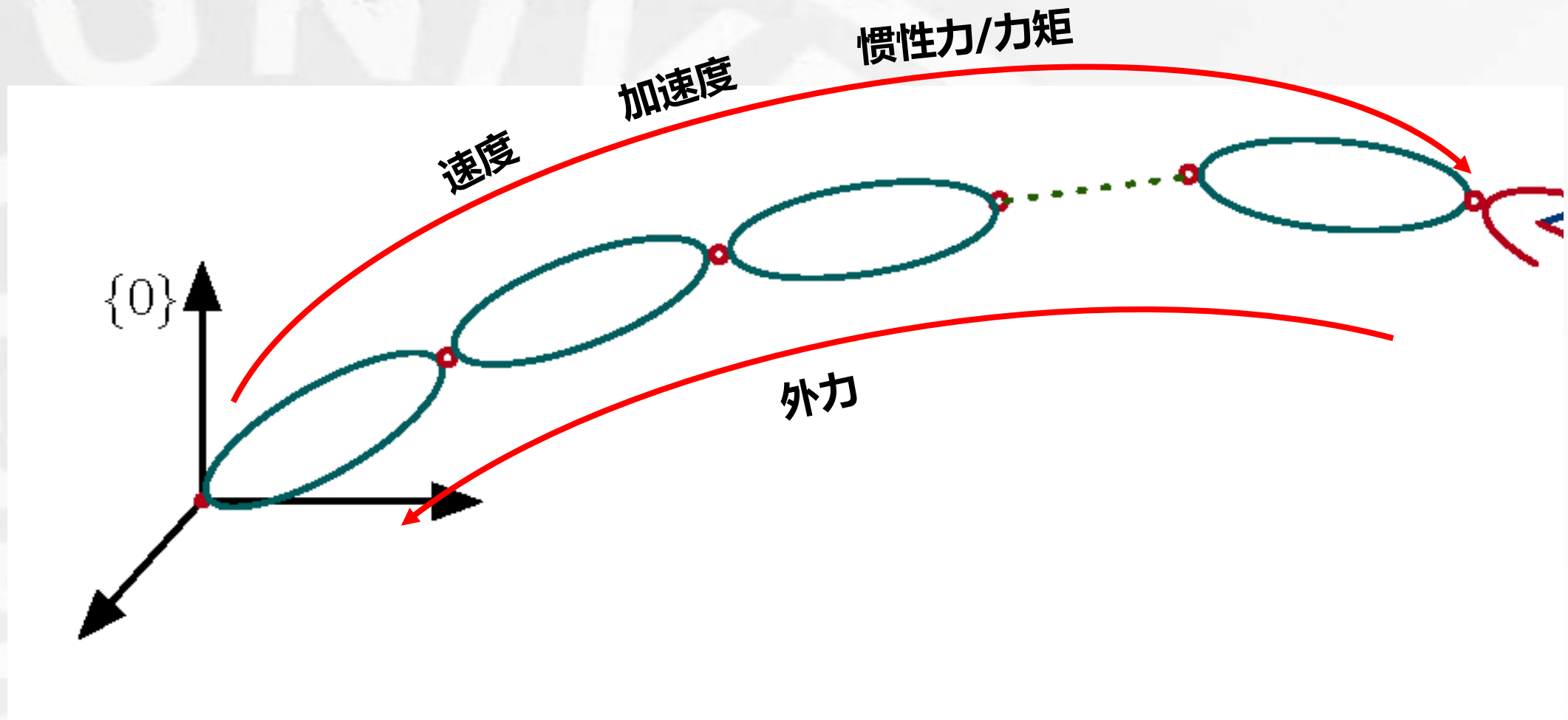
$$F_{i+1} = m_{i+1} \dot{v}_{C_{i+1}}$$

连杆惯性力矩：

$$N_{i+1} = I_{C_{i+1}} \dot{\omega}_{i+1} + \omega_{i+1} \times I_{C_{i+1}} \omega_{i+1}$$



## 2.3 机器人动力学建模过程



## 2.3.3 连杆受力分析

牛顿方程:

$$F_i = m_i \dot{v}_{c_i} = \sum \text{force}$$

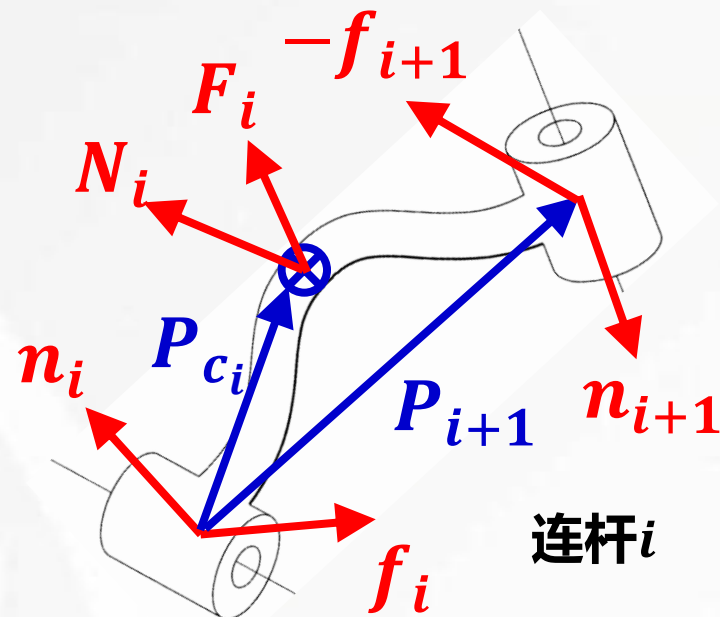
欧拉方程:

$$N_i = I_{c_i} \dot{\omega}_i + \omega_i \times I_{c_i} \omega_i = \sum \text{moments at } c_i$$

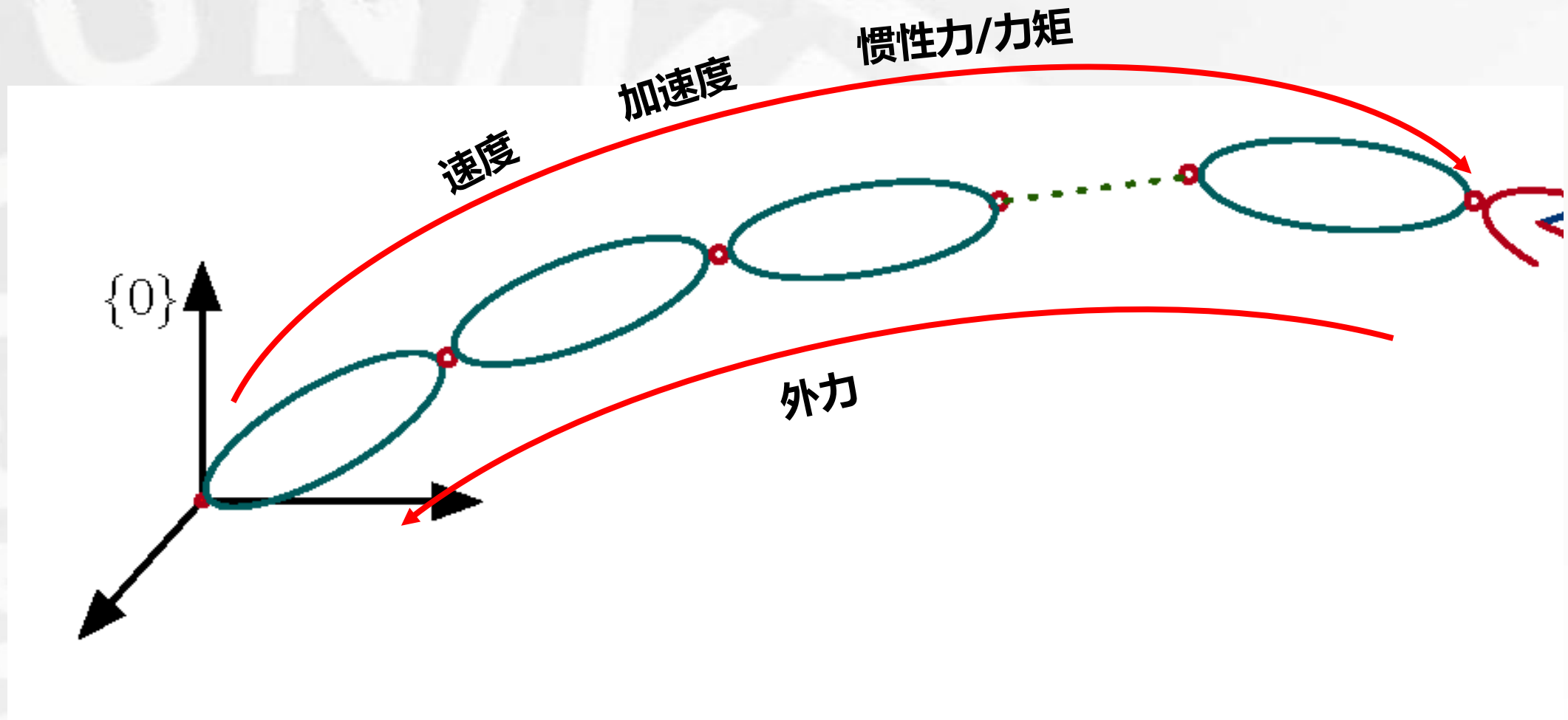
连杆受力分析(质心处):

$$F_i = f_i - f_{i+1} = m_i \dot{v}_{c_i}$$

$$N_i = n_i - n_{i+1} + (-P_{c_i}) \times f_i + (P_{i+1} - P_{c_i}) \times (-f_{i+1})$$



## 2.3 机器人动力学建模过程



## 2.4 牛顿-欧拉动力学建模流程

■ 外推求取速度、加速度、惯性力:  $i: 0 \rightarrow N - 1$

$${}^{i+1}v_{i+1} = {}^{i+1}_i R \ {}^i v_i + {}^{i+1}_i R ({}^i \omega_i \times {}^i P_{i+1}) + \dot{d}_{i+1} \cdot {}^{i+1}Z_{i+1}$$

初始条件:  ${}^0\omega_0, {}^0v_0, {}^0\dot{\omega}_0, {}^0\dot{v}_0$

$${}^{i+1}\omega_{i+1} = {}^{i+1}_i R \ {}^i \omega_i + \dot{\theta}_{i+1} {}^{i+1}Z_{i+1}$$

$${}^{i+1}\dot{\omega}_{i+1} = {}^{i+1}_i R \ {}^i \dot{\omega}_i + \ddot{\theta}_{i+1} {}^{i+1}Z_{i+1} + \dot{\theta}_{i+1} \cdot ({}^{i+1}_i R \ {}^i \omega_i \times {}^{i+1}Z_{i+1})$$

$${}^{i+1}\dot{v}_{i+1} = {}^{i+1}_i R \left( {}^i \dot{v}_i + {}^i \dot{\omega}_i \times {}^i P_{i+1} + {}^i \omega_i \times ({}^i \omega_i \times {}^i P_{i+1}) \right) + \ddot{d}_{i+1} {}^{i+1}Z_{i+1} + 2\dot{d}_{i+1} {}^{i+1}\omega_i \times {}^{i+1}Z_{i+1}$$

$${}^{i+1}\dot{v}_{C_{i+1}} = {}^{i+1}\dot{v}_{i+1} + {}^{i+1}\dot{\omega}_{i+1} \times {}^{i+1}P_{C_{i+1}} + {}^{i+1}\omega_{i+1} \times ({}^{i+1}\omega_{i+1} \times {}^{i+1}P_{C_{i+1}})$$

$${}^{i+1}F_{i+1} = m_{i+1} {}^{i+1}\dot{v}_{C_{i+1}}$$

$${}^{i+1}N_{i+1} = {}^{i+1}I_{C_{i+1}} {}^{i+1}\dot{\omega}_{i+1} + {}^{i+1}\omega_{i+1} \times {}^{i+1}I_{C_{i+1}} {}^{i+1}\omega_{i+1}$$

若机器人基座固定于地面:

$${}^0\dot{v}_0 = [0 \quad 0 \quad g]^T$$

■ 内推求取关节力矩:  $i: N \rightarrow 1$

$${}^i f_i = {}^i F_i + {}_{i+1}^i R {}^{i+1} f_{i+1}$$

$${}^i n_i = {}^i N_i + {}_{i+1}^i R {}^{i+1} n_{i+1} + {}^i P_{C_i} \times {}^i f_i + ({}^i P_{i+1} - {}^i P_{C_i}) \times {}_{i+1}^i R {}^{i+1} f_{i+1}$$

$$\tau_i = \begin{cases} n_i^T \cdot Z_i & \text{旋转关节} \\ f_i^T \cdot Z_i & \text{平移关节} \end{cases}$$

## 2.5 例题1

如图，一RP平面机器人，连杆质量分别为 $m_1, m_2$ ，连杆质心距关节1的距离分别为 $l_1, d_2$ ，连杆其质心处的惯性张量分别为：

$${}^{C_1}I_1 = \begin{bmatrix} I_{xx1} & 0 & 0 \\ 0 & I_{yy1} & 0 \\ 0 & 0 & I_{zz1} \end{bmatrix}, {}^{C_2}I_2 = \begin{bmatrix} I_{xx2} & 0 & 0 \\ 0 & I_{yy2} & 0 \\ 0 & 0 & I_{zz2} \end{bmatrix}$$

求此操作臂的动力学方程。

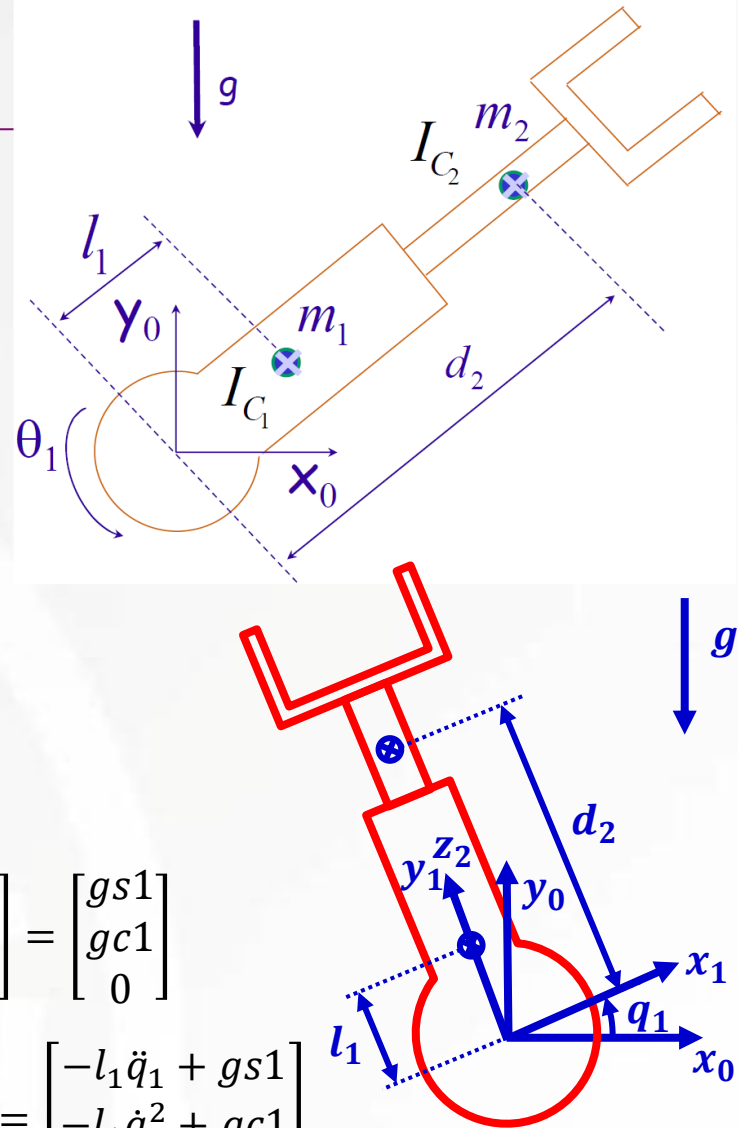
- 初始条件： ${}^0\omega_0 = 0, {}^0v_0 = 0, {}^0\dot{\omega}_0 = 0, {}^0\dot{v}_0 = [0 \ g \ 0]^T, f_3 = 0, n_3 = 0$
- 机器人关节空间坐标： $q = [q_1 \ q_2]^T$ ，广义关节力矩矢量： $\Gamma = [\tau_1 \ \tau_2]^T$
- 外推求取速度、加速度、惯性力： $i: 0 \rightarrow N - 1$

$${}^1\omega_1 = \dot{q}_1 {}^1Z_1 = [0 \ 0 \ \dot{q}_1]^T, {}^1\dot{\omega}_1 = \ddot{q}_1 {}^1Z_1 = [0 \ 0 \ \ddot{q}_1]^T, {}^1\dot{v}_1 = {}^1R({}^0\dot{v}_0) = \begin{bmatrix} c1 & s1 & 0 \\ -s1 & c1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ g \\ 0 \end{bmatrix} = \begin{bmatrix} gs1 \\ gc1 \\ 0 \end{bmatrix}$$

$${}^1\dot{v}_{C_1} = {}^1\dot{v}_1 + {}^1\dot{\omega}_1 \times {}^1P_{C_1} + {}^1\omega_1 \times ({}^1\omega_1 \times {}^1P_{C_1}) = \begin{bmatrix} gs1 \\ gc1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \ddot{q}_1 \end{bmatrix} \times \begin{bmatrix} 0 \\ l_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{q}_1 \end{bmatrix} \times \left( \begin{bmatrix} 0 \\ 0 \\ \dot{q}_1 \end{bmatrix} \times \begin{bmatrix} 0 \\ l_1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} -l_1\ddot{q}_1 + gs1 \\ -l_1\dot{q}_1^2 + gc1 \\ 0 \end{bmatrix}$$

$${}^1F_1 = m_1 {}^1\dot{v}_{C_1}, {}^1N_1 = {}^0R^T {}^{C_1}I_1 {}^0R {}^1\dot{\omega}_1 + {}^1\omega_1 \times {}^0R^T {}^{C_1}I_1 {}^0R {}^1\omega_1 = [0 \ 0 \ \ddot{q}_1 I_{zz1}]^T$$

$${}^{i+1}\dot{v}_{i+1} = {}^{i+1}R \left( {}^i\dot{v}_i + {}^i\dot{\omega}_i \times {}^iP_{i+1} + {}^i\omega_i \times ({}^i\omega_i \times {}^iP_{i+1}) \right) + \ddot{d}_{i+1} {}^{i+1}Z_{i+1} + 2\dot{d}_{i+1} {}^{i+1}\omega_i \times {}^{i+1}Z_{i+1}$$



## 2.5 例题1

■ 初始条件:  ${}^0\omega_0 = 0, {}^0v_0 = 0, {}^0\dot{\omega}_0 = 0, {}^0\dot{v}_0 = [0 \ g \ 0]^T, f_3 = 0, n_3 = 0$

■ 外推求取速度、加速度、惯性力:  $i: 0 \rightarrow N - 1$

$${}^1\omega_1 = [0 \ 0 \ \dot{q}_1]^T, {}^1\dot{\omega}_1 = [0 \ 0 \ \ddot{q}_1]^T, {}^1\dot{v}_1 = [gs1 \ gc1 \ 0]^T, {}^1\dot{v}_{C_1} = [-l_1\ddot{q}_1 + gs1 \ -l_1\dot{q}_1^2 + gc1 \ 0]^T$$

$${}^1F_1 = [-m_1l_1\ddot{q}_1 + m_1gs1 \ -m_1l_1\dot{q}_1^2 + m_1gc1 \ 0]^T, {}^1N_1 = [0 \ 0 \ \ddot{q}_1I_{zz1}]^T$$

$${}^1\omega_2 = {}^1\omega_1 = [0 \ 0 \ \dot{q}_1]^T, {}^1\dot{\omega}_2 = [0 \ 0 \ \ddot{q}_1]^T, {}^1P_2 = [0 \ q_2 \ 0]^T, {}^1P_{C_2} = [0 \ d_2 \ 0]^T$$

$${}^1\dot{v}_2 = {}^1\dot{v}_1 + {}^1\dot{\omega}_1 \times {}^1P_2 + {}^1\omega_1 \times ({}^1\omega_1 \times {}^1P_2) + \ddot{q}_2 {}^1Z_2 + 2\dot{q}_2 {}^1\omega_1 \times {}^1Z_2 = [-\ddot{q}_1q_2 - 2\dot{q}_1\dot{q}_2 + gs1 \ -\dot{q}_1^2q_2 + \ddot{q}_2 + gc1 \ 0]^T$$

$${}^1\dot{v}_{C_2} = {}^1\dot{v}_2 + {}^1\dot{\omega}_2 \times {}^1P_{C_2} + {}^1\omega_2 \times ({}^1\omega_2 \times {}^1P_{C_2}) = [-2\dot{q}_1\dot{q}_2 + gs1 - \ddot{q}_1(d_2 + q_2) \ \ddot{q}_2 + gc1 - \dot{q}_1^2(d_2 + q_2) \ 0]^T$$

$${}^1F_2 = m_2 {}^1\dot{v}_{C_2} = [-2m_2\dot{q}_1\dot{q}_2 + m_2gs1 - m_2\ddot{q}_1(d_2 + q_2) \ m_2\ddot{q}_2 + m_2gc1 - m_2\dot{q}_1^2(d_2 + q_2) \ 0]^T$$

$${}^1N_2 = {}^0R^T C_2 I_2 {}^0R {}^1\dot{\omega}_2 + {}^1\omega_2 \times {}^0R^T C_2 I_2 {}^0R {}^1\omega_2 = [0 \ 0 \ \ddot{q}_1I_{zz2}]^T$$

■ 内推求取关节力矩:  $i: N \rightarrow 1$

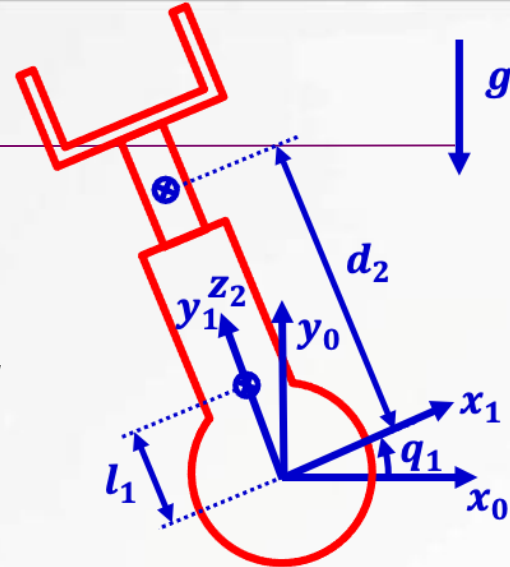
$${}^1f_2 = {}^1F_2, \quad \tau_2 = {}^1f_2^T {}^1Z_2 = m_2 \ddot{q}_2 + m_2gc1 - m_2\dot{q}_1^2(d_2 + q_2)$$

$${}^1n_2 = {}^1N_2 + {}^1P_{C_2} \times {}^1F_2 = [0 \ 0 \ \ddot{q}_1I_{zz2} + 2m_2d_2\dot{q}_1\dot{q}_2 - m_2d_2gs1 + m_2d_2\ddot{q}_1(d_2 + q_2)]^T$$

$${}^1f_1 = {}^1F_1 + {}^1f_2$$

$${}^1n_1 = {}^1N_1 + {}^1n_2 + {}^1P_{C_1} \times {}^1F_1 + {}^1P_2 \times {}^1f_2 \quad \tau_1 = {}^1n_1^T \cdot {}^1Z_1$$

$$= [* \ * \ \ddot{q}_1(I_{zz2} + I_{zz1} + m_1l_1^2 + m_2(d_2 + q_2)^2) + 2m_2\dot{q}_1\dot{q}_2(d_2 + q_2) - (m_1l_1 + m_2(d_2 + q_2))gs1]^T$$





## 2.5 例题1

$$\tau_2 = m_2 \ddot{q}_2 + m_2 g c1 - m_2 \dot{q}_1^2 (d_2 + q_2)$$

$$\tau_1 = \ddot{q}_1 (I_{zz2} + I_{zz1} + m_1 l_1^2 + m_2 (d_2 + q_2)^2) + 2m_2 \dot{q}_1 \dot{q}_2 (d_2 + q_2) - (m_1 l_1 + m_2 (d_2 + q_2)) g s1$$

■ 机器人关节空间状态:

$$q = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}, \dot{q} = \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}, \ddot{q} = \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{bmatrix}$$

■ 机器人广义关节力矩 $\Gamma$ :  $\Gamma = [\tau_1 \quad \tau_2]^T$

■ 机器人动力学方程标准形式:

$$\begin{bmatrix} I_{zz2} + I_{zz1} + m_1 l_1^2 + m_2 (d_2 + q_2)^2 & 0 \\ 0 & m_2 \end{bmatrix} \ddot{q} + \begin{bmatrix} 2m_2 (d_2 + q_2) \dot{q}_1 \dot{q}_2 \\ -m_2 (d_2 + q_2) \dot{q}_1^2 \end{bmatrix} + \begin{bmatrix} -(m_1 l_1 + m_2 (d_2 + q_2)) s1 \\ m_2 c1 \end{bmatrix} g = \Gamma$$

03

# 拉格朗日动力学方程

Lagrange Dynamics Algorithm

# 3.1 拉格朗日动力学建模思想

## ■ 从能量角度对系统进行分析：等价于牛顿力学

□ 系统能量包含动能和势能

□ 系统能量变化：外力做功

□ 势能 $U$ ：弹性势能和重力势能

- ✓ 势能仅与系统的位置相关，与速度无关  $\frac{\partial U}{\partial \dot{q}} = 0$
- ✓ 势能的变化：克服弹性元件或重力做功
- ✓ 单位位移下势能的变化：克服弹性力或重力  $\frac{\partial U}{\partial q} = G(q)$

□ 动能 $K$ ：

- ✓ 动能与系统的速度与系统构型相关，与位置无关

- ✓ 动能的变化：

- 速度变化带来的动能变化：动量或角动量  $\frac{\partial K}{\partial \dot{q}} = Q(q, \dot{q})$
- (角)动量守恒定理：(角)动量的变化等于系统受的惯性力  $F = \frac{d}{dt} Q(q, \dot{q})$
- 构型变化带来的动能变化：系统内力做功  $\frac{\partial K}{\partial q} = \frac{\partial M(q)}{\partial q}$

系统输入 $\Gamma$ ：

$$\begin{aligned}\Gamma &= \frac{d}{dt} \frac{\partial K}{\partial \dot{q}} + \frac{\partial U}{\partial q} - \frac{\partial K}{\partial q} \\ &= \frac{d}{dt} \frac{\partial (K - U)}{\partial \dot{q}} - \frac{\partial (K - U)}{\partial q} \\ &= \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q}\end{aligned}$$

## 3.2 欧拉-拉格朗日动力学方程

- 对于具有外力作用的非保守机械系统，其拉格朗日动力函数 $\mathcal{L}$ 可定义为：

$$\mathcal{L}(q, \dot{q}, t) = K(q, \dot{q}, t) - U(q, t)$$

其中：  $K(q, \dot{q}, t)$ 为系统总动能，  $U(q, t)$ 为系统总势能，  $q \in \mathbb{R}^n$  为广义坐标，描述系统的构型，  $\dot{q} \in \mathbb{R}^n$ 为广义速度，且广义坐标和广义速度是相互独立的变量（确定系统初始状态，观测的初始时刻，位置和速度是相互独立的）。

- 带外力的欧拉-拉格朗日方程：

$$f = \frac{d}{dt} \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial \dot{q}} - \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial q}$$

其中：  $f \in \mathbb{R}^n$  为广义力，且与广义速度 $\dot{q}$ 是对偶的，他们的内积 $f^T \dot{q}$ 对应于功率。

## 3.3 例1 拉格朗日动力学建模

■ 系统动能 $K$ :

$$K = \frac{1}{2} m \dot{x}^2$$

■ 系统势能 $U$ :

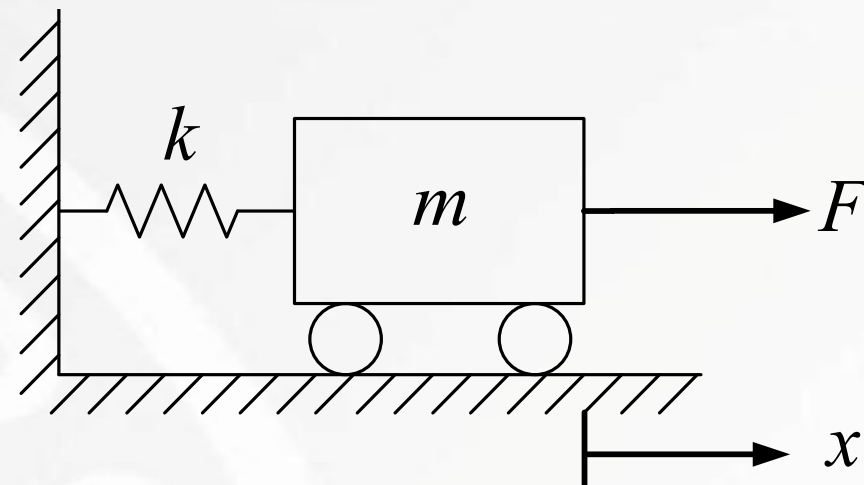
$$U = \frac{1}{2} k x^2$$

■ 拉格朗日函数 $\mathcal{L}$ :

$$\mathcal{L} = K - U$$

■ 拉格朗日动力学方程:

$$F = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} - \frac{\partial \mathcal{L}}{\partial x} = \frac{d}{dt} (m\dot{x} - 0) - (0 - kx) = m\ddot{x} + kx \Rightarrow m\ddot{x} = F - kx$$



## 3.4 机器人拉格朗日动力学方程

- 对于具有 $N$ 个自由度的机器人， $q \in \mathbb{R}^N$  为机器人广义构型状态， $\dot{q} \in \mathbb{R}^N$  为机器人广义关节速度， $\Gamma \in \mathbb{R}^N$  为机器人广义力矩，其拉格朗日动力函数 $\mathcal{L}$ 可定义为：

$$\mathcal{L}(q, \dot{q}, t) = K(q, \dot{q}, t) - U(q, t)$$

其中： $K(q, \dot{q}, t)$ 为机器人系统总动能， $U(q, t)$ 为机器人系统总势能。

- 机器人动力学方法可有如下公式获得：

$$\Gamma = \frac{d}{dt} \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial \dot{q}} - \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial q}$$

其中：

$$\tau_i = \frac{d}{dt} \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial q_i} \quad i = 1, 2, \dots, N$$

$$\tau_i = \begin{cases} n_i^T \cdot Z_i & \text{旋转关节} \\ f_i^T \cdot Z_i & \text{平移关节} \end{cases}$$

### 3.4.1 机器人拉格朗日动力学方程组成

$$\Gamma = \frac{d}{dt} \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial \dot{q}} - \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial q}$$

$$\mathcal{L}(q, \dot{q}, t) = K(q, \dot{q}, t) - U(q, t)$$

其中：  $K(q, \dot{q}, t)$  为机器人系统总动能，  $U(q, t)$  为机器人系统总势能。

$$\Gamma = \frac{d}{dt} \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial \dot{q}} - \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial q} = \underbrace{\frac{d}{dt} \frac{\partial K(q, \dot{q}, t)}{\partial \dot{q}} - \frac{\partial K(q, \dot{q}, t)}{\partial q}}_{\text{惯性力}} + \underbrace{\frac{\partial U(q, t)}{\partial q}}_{\text{重力}}$$

$$\boxed{M(q)\ddot{q} + V(q, \dot{q})} + \boxed{G(q)} = \Gamma$$



## 3.4.2 机器人拉格朗日动力学方程推导过程

$$\begin{aligned} K(q, \dot{q}) &= \frac{1}{2} \dot{q}^T M(q) \dot{q} \Rightarrow \frac{\partial}{\partial \dot{q}} K(q, \dot{q}) = \frac{\partial}{\partial \dot{q}} \left( \frac{1}{2} \dot{q}^T M(q) \dot{q} \right) = M(q) \dot{q} \\ &\Rightarrow \frac{d}{dt} \left( \frac{\partial}{\partial \dot{q}} K(q, \dot{q}) \right) = \mathbf{M}(q) \ddot{q} + \dot{\mathbf{M}}(q) \dot{q} \\ K(q, \dot{q}) &= \frac{1}{2} \dot{q}^T M(q) \dot{q} \Rightarrow \frac{\partial}{\partial q} K(q, \dot{q}) = \frac{\partial}{\partial q} \left( \frac{1}{2} \dot{q}^T M(q) \dot{q} \right) = \frac{1}{2} \begin{bmatrix} \dot{q}^T \frac{\partial M(q)}{\partial q_1} \dot{q} \\ \vdots \\ \dot{q}^T \frac{\partial M(q)}{\partial q_N} \dot{q} \end{bmatrix} \\ \frac{d}{dt} \frac{\partial K(q, \dot{q}, t)}{\partial \dot{q}} - \frac{\partial K(q, \dot{q}, t)}{\partial q} &= \mathbf{M}(q) \ddot{q} + \dot{\mathbf{M}}(q) \dot{q} - \frac{1}{2} \begin{bmatrix} \dot{q}^T \frac{\partial M(q)}{\partial q_1} \dot{q} \\ \vdots \\ \dot{q}^T \frac{\partial M(q)}{\partial q_N} \dot{q} \end{bmatrix} \\ \mathbf{M}(q) \ddot{q} + \mathbf{V}(q, \dot{q}) + G(q) &= \Gamma \end{aligned}$$

## 3.4.2 机器人拉格朗日动力学方程推导过程

- 为了求解如下所示的机器人动力学方程：

$$M(q)\ddot{q} + V(q, \dot{q}) + G(q) = \Gamma$$

- 需要求解  $M(q)$ ,  $V(q, \dot{q})$ ,  $G(q)$
- $V(q, \dot{q})$  可由  $M(q)$  求得

$$V(q, \dot{q}) = \dot{M}(q)\dot{q} - \frac{1}{2} \begin{bmatrix} \dot{q}^T \frac{\partial M(q)}{\partial q_1} \dot{q} \\ \vdots \\ \dot{q}^T \frac{\partial M(q)}{\partial q_N} \dot{q} \end{bmatrix}$$

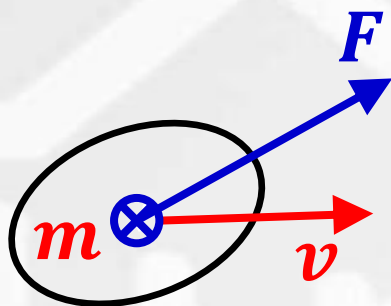
- 如可计算得到系统动能  $K(q, \dot{q})$ ，则可求取  $M(q)$

$$M(q): K(q, \dot{q}) = \frac{1}{2} \dot{q}^T M(q) \dot{q}$$

## 3.5.1 刚体动能

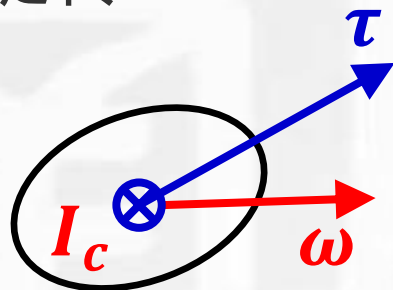
动能是指力在一个过程中对物体所做的功等于在这个过程中动能的变化。具有瞬时性，动能是**标量**，**无负值**。

■ 平移：



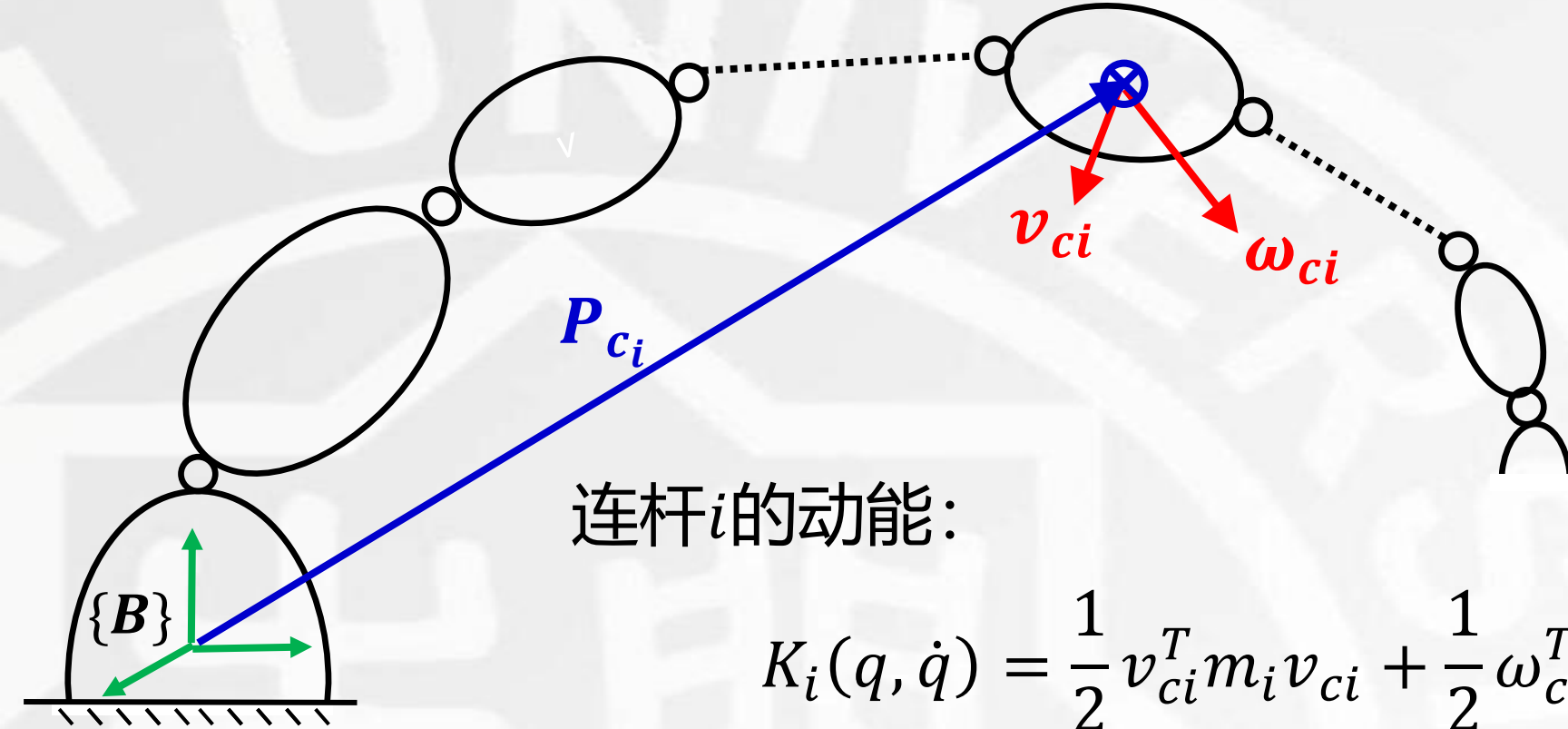
$$K(q, \dot{q}) = \frac{1}{2} v^T m v$$

■ 旋转：



$$K(q, \dot{q}) = \frac{1}{2} \omega^T I_c \omega$$

## 3.5.2 机器人系统动能求取——多体动力学



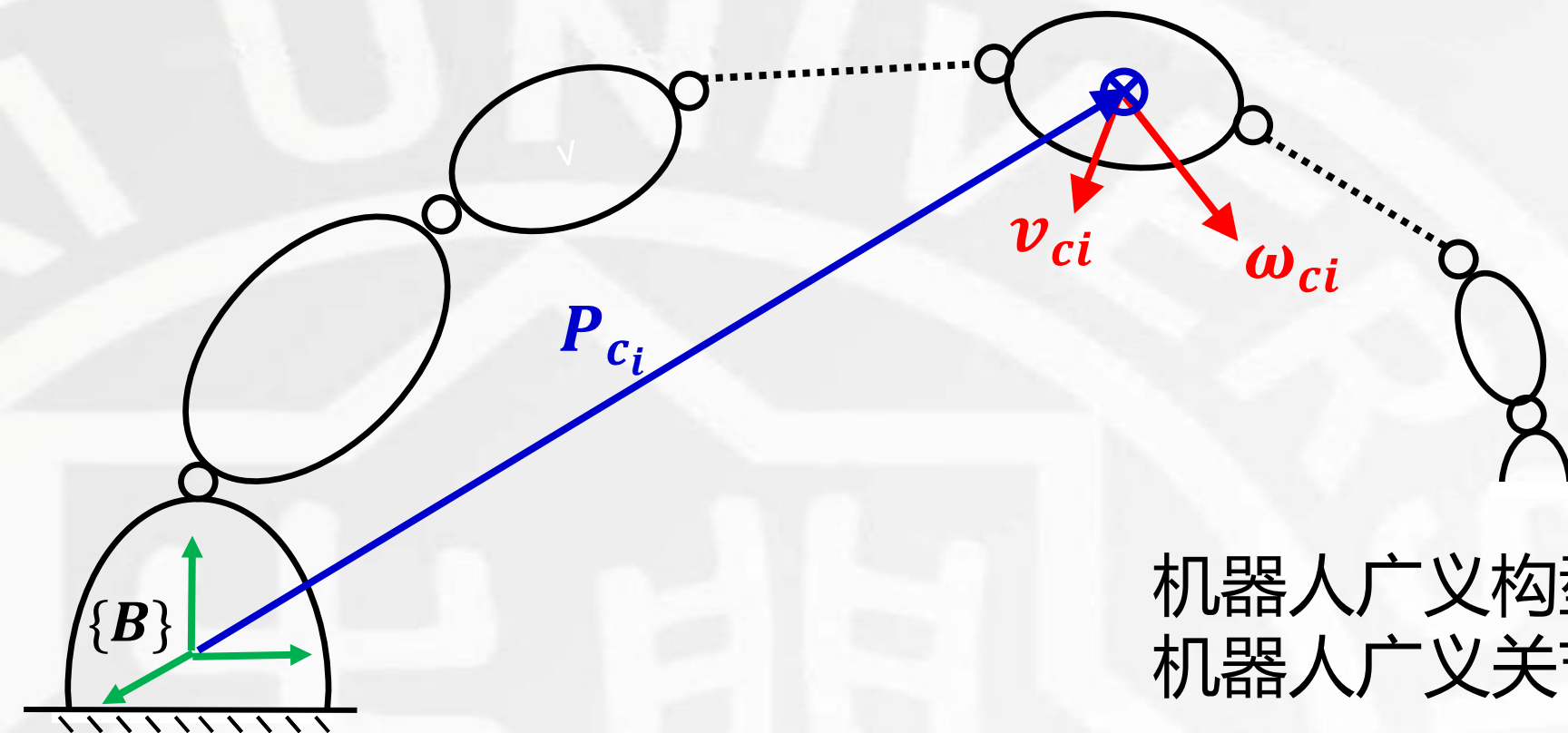
连杆 $i$ 的动能:

$$K_i(q, \dot{q}) = \frac{1}{2} v_{ci}^T m_i v_{ci} + \frac{1}{2} \omega_{ci}^T I_{ci} \omega_{ci}$$

机器人系统总动能:

$$K(q, \dot{q}) = \sum_{i=1}^n K_i = \sum_{i=1}^n \left( \frac{1}{2} v_{ci}^T m_i v_{ci} + \frac{1}{2} \omega_{ci}^T I_{ci} \omega_{ci} \right)$$

### 3.5.3 机器人系统动能一般形式

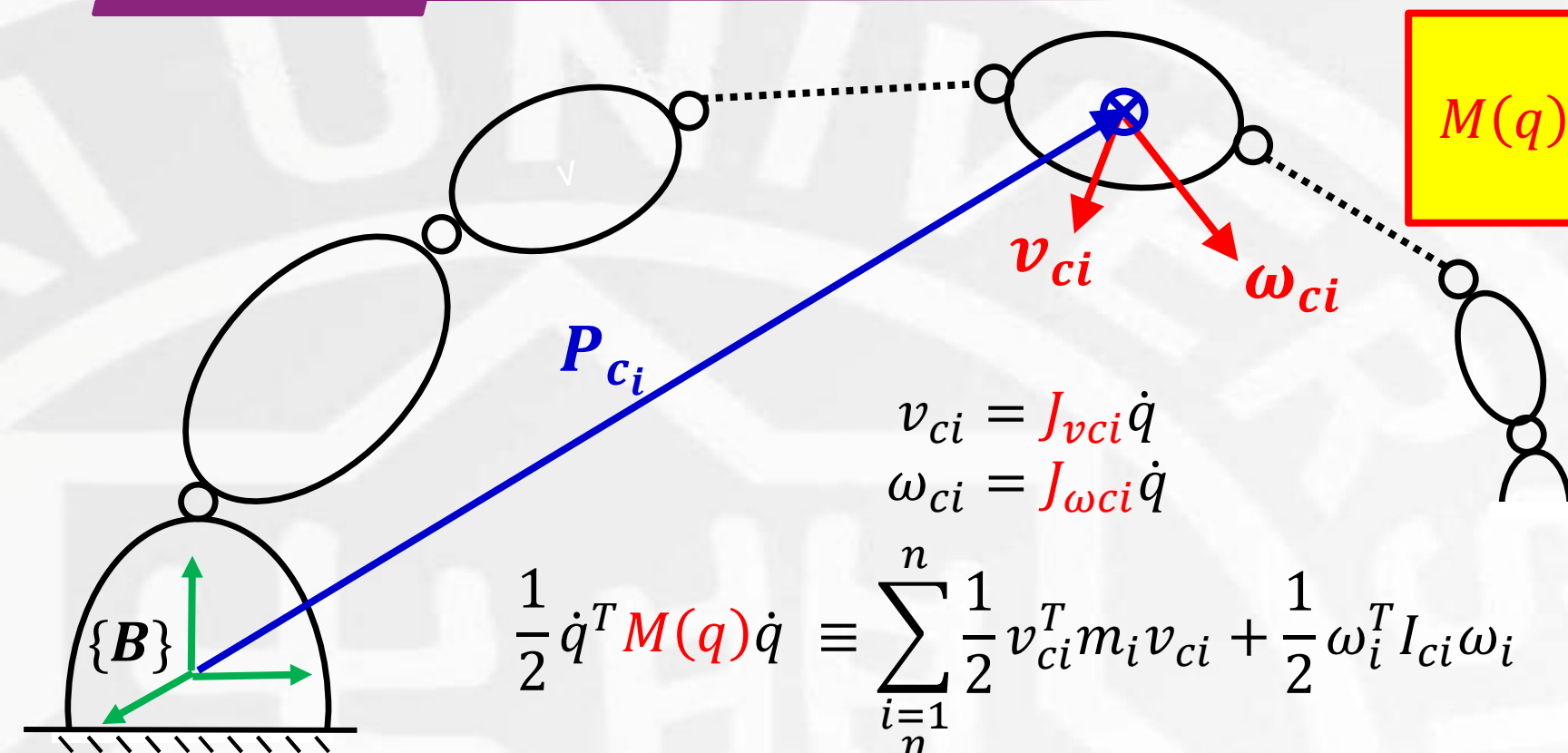


机器人广义构型状态:  $\mathbf{q} \in \mathbb{R}^N$   
机器人广义关节速度:  $\dot{\mathbf{q}} \in \mathbb{R}^N$

机器人系统总动能: 机器人广义关节速度的二次型

$$K(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}} \equiv \sum_{i=1}^n \frac{1}{2} v_{ci}^T m_i v_{ci} + \frac{1}{2} \omega_{ci}^T I_{ci} \omega_{ci}$$

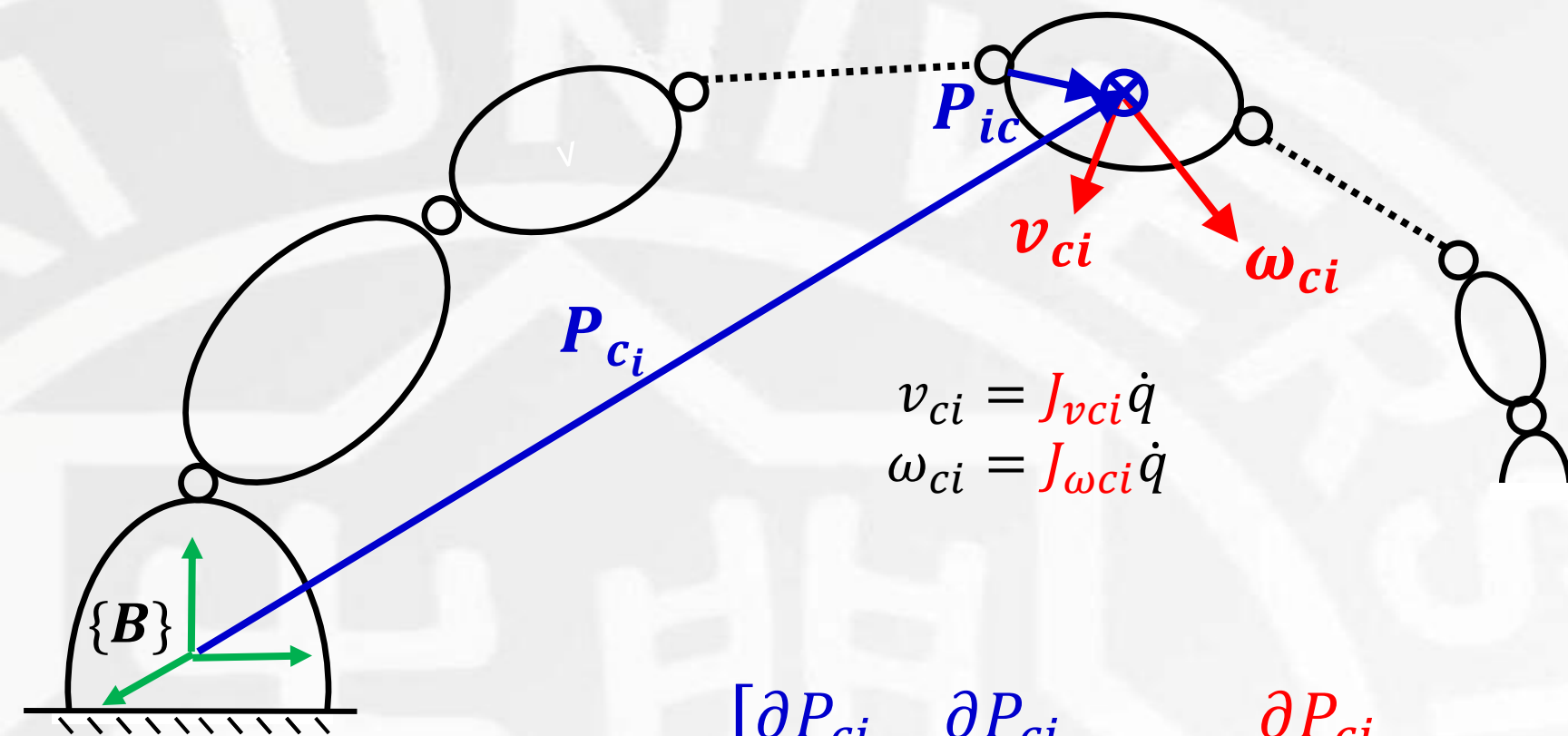
### 3.5.4 机器人系统惯性阵求取



$$M(q) = \sum_{i=1}^n J_{vci}^T m_i J_{vci} + J_{\omega ci}^T I_{ci} J_{\omega ci}$$

$$\begin{aligned} \frac{1}{2} \dot{q}^T M(q) \dot{q} &\equiv \sum_{i=1}^n \frac{1}{2} v_{ci}^T m_i v_{ci} + \frac{1}{2} \omega_i^T I_{ci} \omega_i \\ &= \sum_{i=1}^n \frac{1}{2} \dot{q}^T J_{vci}^T m_i J_{vci} \dot{q} + \frac{1}{2} \dot{q}^T J_{\omega ci}^T I_{ci} J_{\omega ci} \dot{q} \\ &= \frac{1}{2} \dot{q}^T \left[ \sum_{i=1}^n J_{vci}^T m_i J_{vci} + J_{\omega ci}^T I_{ci} J_{\omega ci} \right] \dot{q} \end{aligned}$$

### 3.5.4 机器人系统惯性阵求取



$$\begin{aligned} v_{ci} &= J_{vci} \dot{q} \\ \omega_{ci} &= J_{\omega ci} \dot{q} \end{aligned}$$

$$J_{ci} = \begin{bmatrix} I & -\hat{P}_{ic} \\ 0 & I \end{bmatrix} J_i$$

$$J_{vci} = \begin{bmatrix} \frac{\partial P_{ci}}{\partial q_1} & \frac{\partial P_{ci}}{\partial q_2} & \cdots & \frac{\partial P_{ci}}{\partial q_i} & 0 & 0 & \cdots & 0 \end{bmatrix}$$

$$J_{\omega ci} = [\bar{\varepsilon}_1 Z_1 \quad \bar{\varepsilon}_2 Z_2 \quad \cdots \quad \bar{\varepsilon}_i Z_i \quad 0 \quad 0 \quad \cdots \quad 0]$$



## 3.5.5 惯性阵性质

$$M(q) = \begin{bmatrix} \boxed{m_{11}} & m_{12} & \cdots & m_{1n-1} & m_{1n} \\ m_{21} & \boxed{m_{22}} & \cdots & m_{2n-1} & m_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ m_{n-1,1} & m_{n-1,2} & \cdots & m_{n-1n-1} & m_{n-1n} \\ m_{n1} & m_{n2} & \cdots & m_{nn-1} & \boxed{m_{nn}} \end{bmatrix}_{n \times n}$$

- 惯性阵  $M(q) \in R^{n \times n}$  为**正定的对称阵**
- 惯性阵  $M(q)$  对角线元素  $m_{ii}$  描述机器人在  $q_{i+1}$  到  $q_n$  的状态决定构型下的惯量，即  $m_{ii}(q_{i+1}, \dots, q_n)$ ，其表征的是关节  $i$  之后的关节(关节  $i + 1$  到关节  $n$ )在当前状态下固定后的(平移或转动) 惯量
  - $m_{nn}$  为定值
  - 机器人系统可以看作从最后一个关节开始组装，每当添加一个关节  $j$ ，就会在惯性阵的最上方增加一行，最左边增加一列，表征的是关节  $j$  运动的加速度与其他关节间的耦合作用

### 3.5.6 惯性力 $V(q, \dot{q})$ 求取(1)

$$V(q, \dot{q}) = \dot{M}(q)\dot{q} - \frac{1}{2} \begin{bmatrix} \dot{q}^T \frac{\partial M(q)}{\partial q_1} \dot{q} \\ \vdots \\ \dot{q}^T \frac{\partial M(q)}{\partial q_n} \dot{q} \end{bmatrix} \quad \dot{M}(q) = \frac{d}{dt} \begin{bmatrix} m_{11} & m_{12} & \cdots & m_{1n} \\ m_{21} & m_{22} & \cdots & m_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ m_{n1} & m_{n2} & \cdots & m_{nn} \end{bmatrix} = \frac{\partial M(q)}{\partial q} \dot{q} = \begin{bmatrix} m'_{11} & m'_{12} & \cdots & m'_{1n} \\ m'_{21} & m'_{22} & \cdots & m'_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ m'_{n1} & m'_{n2} & \cdots & m'_{nn} \end{bmatrix}$$

$$m'_{ij} = \frac{\partial m_{i,j}}{\partial q_1} \dot{q}_1 + \frac{\partial m_{i,j}}{\partial q_2} \dot{q}_2 + \cdots + \frac{\partial m_{i,j}}{\partial q_n} \dot{q}_n$$

$$\dot{M}(q)_{n \times n} \dot{q}_{n \times 1} = [p_1 \quad p_2 \quad \cdots \quad p_n]^T$$

$$p_i = m'_{i1} \dot{q}_1 + m'_{i2} \dot{q}_2 + \cdots + m'_{in} \dot{q}_n = \frac{\partial m_{i,1}}{\partial q_1} \dot{q}_1^2 + \frac{\partial m_{i,1}}{\partial q_2} \dot{q}_1 \dot{q}_2 + \cdots + \frac{\partial m_{i,1}}{\partial q_n} \dot{q}_1 \dot{q}_n \\ + \cdots + \frac{\partial m_{i,n}}{\partial q_1} \dot{q}_1 \dot{q}_n + \frac{\partial m_{i,n}}{\partial q_2} \dot{q}_2 \dot{q}_n + \cdots + \frac{\partial m_{i,n}}{\partial q_n} \dot{q}_n^2$$

令

$$b_{i,j,k} = \frac{\partial m_{i,j}}{\partial q_k}, \quad c_{i,jk} = b_{i,j,k} + b_{i,k,j}$$

$$\dot{M}(q)\dot{q} = \begin{bmatrix} b_{1,1,1} & b_{1,2,2} & \cdots & b_{1,n,n} \\ b_{2,1,1} & b_{2,2,2} & \cdots & b_{2,n,n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n,1,1} & b_{n,2,2} & \cdots & b_{n,n,n} \end{bmatrix}_{n \times n} \begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \\ \vdots \\ \dot{q}_n^2 \end{bmatrix}_n + \begin{bmatrix} c_{1,12} & c_{1,13} & \cdots & c_{1,n-1n} \\ c_{2,12} & c_{2,13} & \cdots & c_{2,n-1n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n,12} & c_{n,13} & \cdots & c_{n,n-1n} \end{bmatrix}_{n \times \frac{(n-1)n}{2}} \begin{bmatrix} \dot{q}_1 \dot{q}_2 \\ \dot{q}_1 \dot{q}_3 \\ \vdots \\ \dot{q}_{n-1} \dot{q}_n \end{bmatrix}_{\frac{(n-1)n}{2}}$$

## 3.5.6 惯性力 $V(q, \dot{q})$ 求取(2)

$$V(q, \dot{q}) = \dot{M}(q)\dot{q} - \frac{1}{2} \begin{bmatrix} \dot{q}^T \frac{\partial M(q)}{\partial q_1} \dot{q} \\ \vdots \\ \dot{q}^T \frac{\partial M(q)}{\partial q_n} \dot{q} \end{bmatrix}$$

$$\frac{\partial M(q)}{\partial q_i} \dot{q} = p_i = [p_{i,1} \quad p_{i,2} \quad \cdots \quad p_{i,n}]^T$$

$$p_{i,j} = \frac{\partial m_{j,1}}{\partial q_i} \dot{q}_1 + \frac{\partial m_{j,2}}{\partial q_i} \dot{q}_2 + \cdots + \frac{\partial m_{j,n}}{\partial q_i} \dot{q}_n$$

$$\dot{q}^T \frac{\partial M(q)}{\partial q_i} \dot{q} = \dot{q}^T p_i = \frac{\partial m_{1,1}}{\partial q_i} \dot{q}_1^2 + \frac{\partial m_{1,2}}{\partial q_i} \dot{q}_1 \dot{q}_2 + \cdots + \frac{\partial m_{1,n}}{\partial q_i} \dot{q}_1 \dot{q}_n$$

$$+ \cdots + \frac{\partial m_{n,1}}{\partial q_i} \dot{q}_1 \dot{q}_n + \frac{\partial m_{n,2}}{\partial q_i} \dot{q}_2 \dot{q}_n + \cdots + \frac{\partial m_{n,n}}{\partial q_i} \dot{q}_n^2$$

令

$$b_{i,j,k} = \frac{\partial m_{i,j}}{\partial q_k}, \quad b_{i,j,k} + b_{j,i,k} = 2b_{i,j,k} \quad ?$$

$$\dot{q}^T \frac{\partial M(q)}{\partial q_i} \dot{q} = [b_{1,1,i} \quad b_{2,2,i} \quad \cdots \quad b_{n,n,i}] \begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \\ \vdots \\ \dot{q}_n^2 \end{bmatrix}_n + [2b_{1,2,i} \quad 2b_{1,3,i} \quad \cdots \quad 2b_{n-1,n,i}] \begin{bmatrix} \dot{q}_1 \dot{q}_2 \\ \dot{q}_1 \dot{q}_3 \\ \vdots \\ \dot{q}_{n-1} \dot{q}_n \end{bmatrix}_{\frac{(n-1)n}{2}}$$

$$\begin{bmatrix} \dot{q}^T \frac{\partial M(q)}{\partial q_1} \dot{q} \\ \vdots \\ \dot{q}^T \frac{\partial M(q)}{\partial q_n} \dot{q} \end{bmatrix} = \begin{bmatrix} b_{1,1,1} & b_{2,2,1} & \cdots & b_{n,n,1} \\ b_{1,1,2} & b_{2,2,2} & \cdots & b_{n,n,2} \\ \vdots & \vdots & \ddots & \vdots \\ b_{1,1,n} & b_{2,2,n} & \cdots & b_{n,n,n} \end{bmatrix}_{n \times n} \begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \\ \vdots \\ \dot{q}_n^2 \end{bmatrix}_n + 2 \begin{bmatrix} b_{1,2,1} & b_{1,3,1} & \cdots & b_{n-1,n,1} \\ b_{1,2,2} & b_{1,3,2} & \cdots & b_{n-1,n,2} \\ \vdots & \vdots & \ddots & \vdots \\ b_{1,2,n} & b_{1,3,n} & \cdots & b_{n-1,n,n} \end{bmatrix}_{n \times \frac{(n-1)n}{2}} \begin{bmatrix} \dot{q}_1 \dot{q}_2 \\ \dot{q}_1 \dot{q}_3 \\ \vdots \\ \dot{q}_{n-1} \dot{q}_n \end{bmatrix}_{\frac{(n-1)n}{2}}$$

### 3.5.6 惯性力 $V(q, \dot{q})$ 求取(3)

$$V(q, \dot{q}) = \dot{M}(q)\dot{q} - \frac{1}{2} \begin{bmatrix} \dot{q}^T \frac{\partial M(q)}{\partial q_1} \dot{q} \\ \vdots \\ \dot{q}^T \frac{\partial M(q)}{\partial q_n} \dot{q} \end{bmatrix} \quad \dot{M}(q)\dot{q} = \begin{bmatrix} b_{1,1,1} & b_{1,2,2} & \cdots & b_{1,n,n} \\ b_{2,1,1} & b_{2,2,2} & \cdots & b_{2,n,n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n,1,1} & b_{n,2,2} & \cdots & b_{n,n,n} \end{bmatrix} \begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \\ \vdots \\ \dot{q}_n^2 \end{bmatrix} + \begin{bmatrix} c_{1,12} & c_{1,13} & \cdots & c_{1,n-1n} \\ c_{2,12} & c_{2,13} & \cdots & c_{2,n-1n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n,12} & c_{n,13} & \cdots & c_{n,n-1n} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \dot{q}_2 \\ \dot{q}_1 \dot{q}_3 \\ \vdots \\ \dot{q}_{n-1} \dot{q}_n \end{bmatrix}$$

$$b_{i,j,k} = \frac{\partial m_{i,j}}{\partial q_k}, \quad c_{i,j,k} = b_{i,j,k} + b_{i,k,j}$$

$$\begin{bmatrix} \dot{q}^T \frac{\partial M(q)}{\partial q_1} \dot{q} \\ \vdots \\ \dot{q}^T \frac{\partial M(q)}{\partial q_n} \dot{q} \end{bmatrix} = \begin{bmatrix} b_{1,1,1} & b_{2,2,1} & \cdots & b_{n,n,1} \\ b_{1,1,2} & b_{2,2,2} & \cdots & b_{n,n,2} \\ \vdots & \vdots & \ddots & \vdots \\ b_{1,1,n} & b_{2,2,n} & \cdots & b_{n,n,n} \end{bmatrix} \begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \\ \vdots \\ \dot{q}_n^2 \end{bmatrix} + 2 \begin{bmatrix} b_{1,2,1} & b_{1,3,1} & \cdots & b_{n-1,n,1} \\ b_{1,2,2} & b_{1,3,2} & \cdots & b_{n-1,n,2} \\ \vdots & \vdots & \ddots & \vdots \\ b_{1,2,n} & b_{1,3,n} & \cdots & b_{n-1,n,n} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \dot{q}_2 \\ \dot{q}_1 \dot{q}_3 \\ \vdots \\ \dot{q}_{n-1} \dot{q}_n \end{bmatrix}$$

$$b_{i,j,k} = \frac{\partial m_{i,j}}{\partial q_k}, \quad c_{ijk} = \frac{1}{2} (b_{i,j,k} + b_{i,k,j} - b_{j,k,i})$$

离心力

$$b_{i,j,k} = \frac{\partial m_{i,j}}{\partial q_k}, \quad c_{ij,k} = b_{i,j,k} + b_{j,i,k}$$

科里奥利力

$$V(q, \dot{q}) = \underbrace{\begin{bmatrix} c_{111} & c_{122} & \cdots & c_{1nn} \\ c_{211} & c_{222} & \cdots & c_{2nn} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n11} & c_{n22} & \cdots & c_{nnn} \end{bmatrix}}_{C(q)} \underbrace{\begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \\ \vdots \\ \dot{q}_n^2 \end{bmatrix}}_{[\dot{q}^2]} + \underbrace{\begin{bmatrix} 2c_{112} & 2c_{113} & \cdots & 2c_{1(n-1)n} \\ 2c_{212} & 2c_{213} & \cdots & 2c_{2(n-1)n} \\ \vdots & \vdots & \ddots & \vdots \\ 2c_{n12} & 2c_{n13} & \cdots & 2c_{n(n-1)n} \end{bmatrix}}_{B(q)} \underbrace{\begin{bmatrix} \dot{q}_1 \dot{q}_2 \\ \dot{q}_1 \dot{q}_3 \\ \vdots \\ \dot{q}_{n-1} \dot{q}_n \end{bmatrix}}_{[\dot{q}\dot{q}]}$$

## 3.6 势能

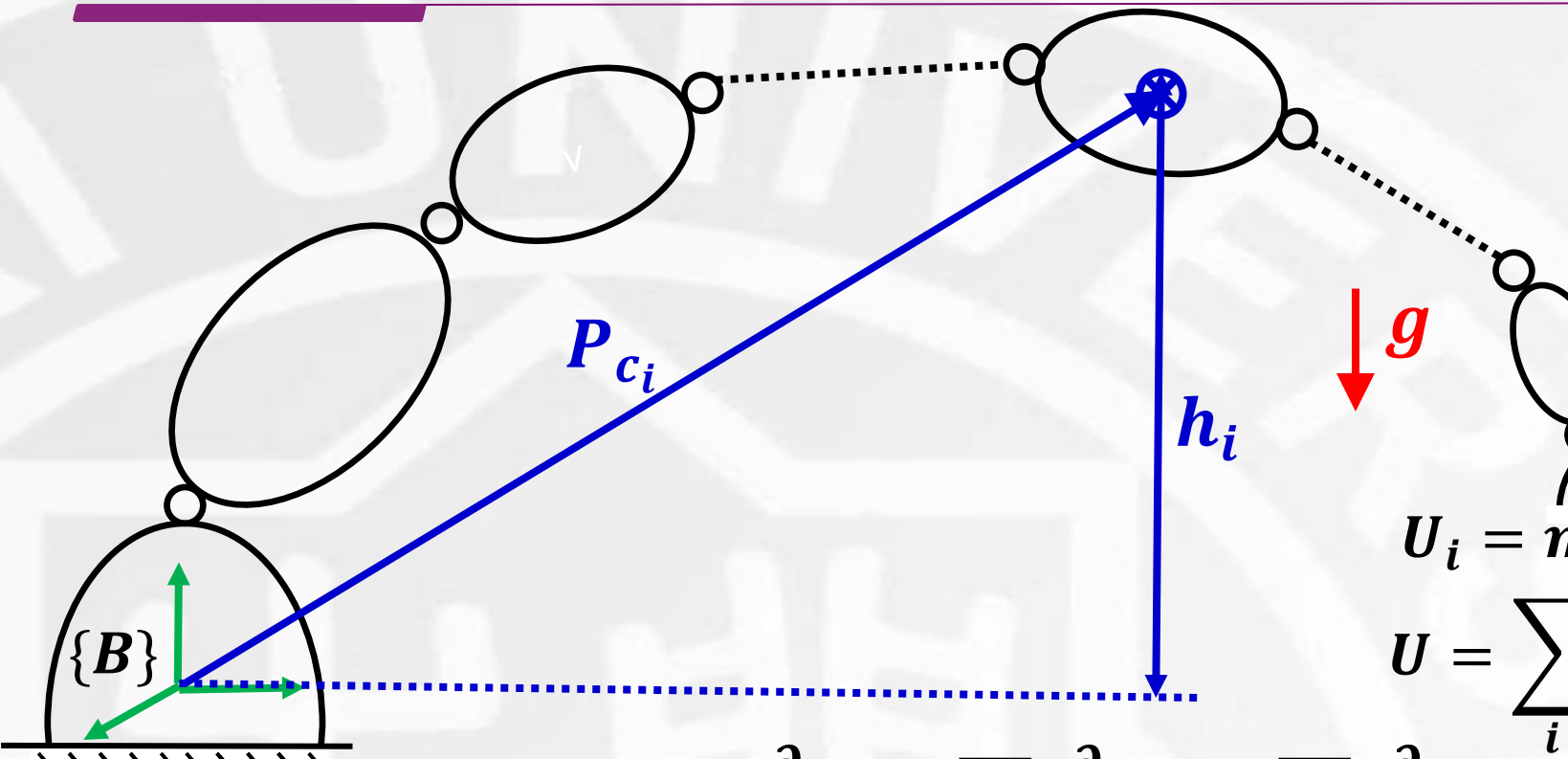


Diagram illustrating the calculation of potential energy for a robotic arm. The diagram shows a base coordinate frame  $\{B\}$  and a link with center of mass  $P_{ci}$ . The height  $h_i$  is measured from a horizontal dashed line to the center of mass. Gravity  $g$  acts downwards.

The potential energy  $U_i$  for a link is given by:

$$U_i = m_i g h_i + U_0 = m_i (-P_{ci}^T g)$$

The total potential energy  $U$  is the sum of the potential energies of all links:

$$U = \sum_i U_i = \sum_i m_i (-P_{ci}^T g)$$

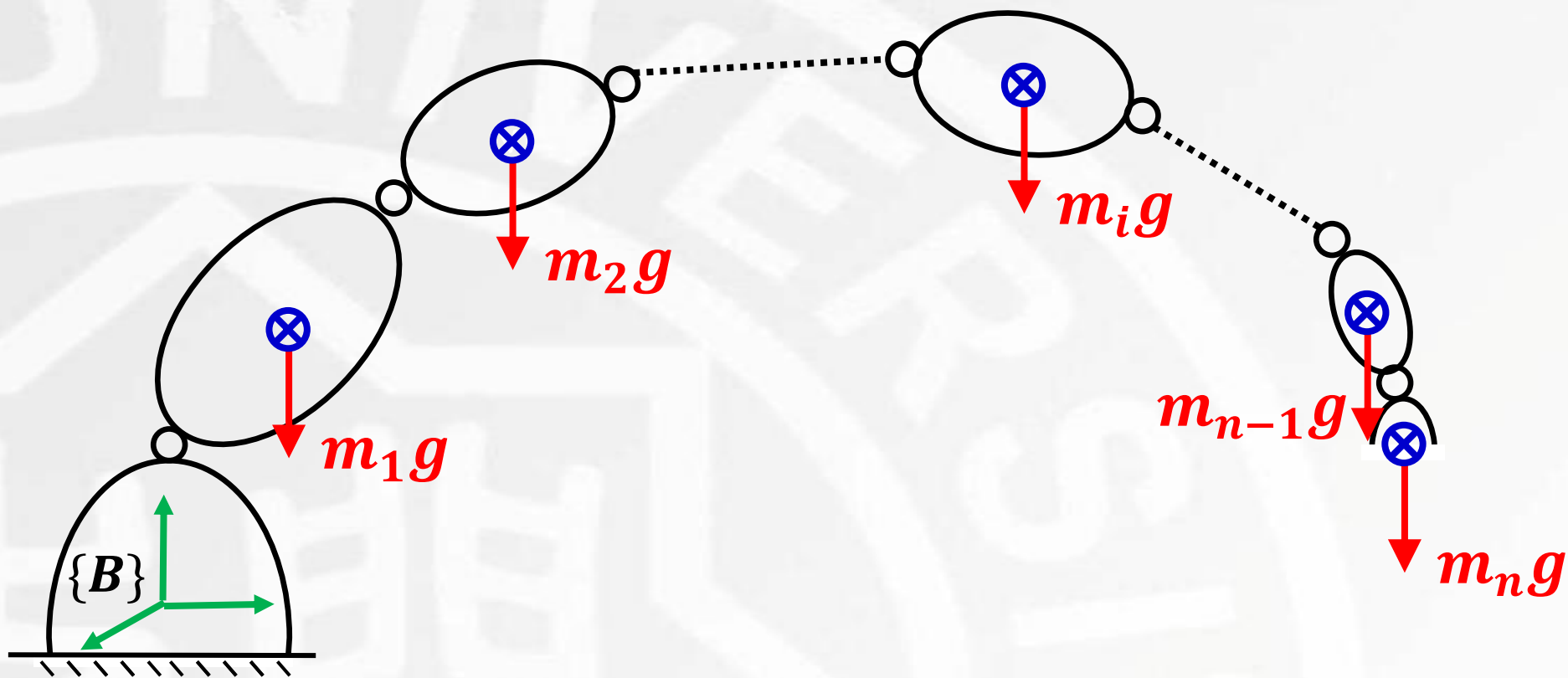
The generalized force  $G(q)$  is the derivative of the total potential energy with respect to the generalized coordinates  $q$ :

$$G(q) = \frac{\partial}{\partial q} U = \sum_i \frac{\partial}{\partial q} U_i = \sum_i \frac{\partial}{\partial q} m_i (-P_{ci}^T g) = - \sum_i m_i \left( \frac{\partial P_{ci}}{\partial q} \right)^T g$$

The generalized force  $G(q)$  can be expressed as a matrix product:

$$G(q) = - [J_{vc1}^T \quad J_{vc2}^T \quad \cdots \quad J_{vcn}^T] \begin{bmatrix} m_1 g \\ m_2 g \\ \vdots \\ m_n g \end{bmatrix}$$

## 3.6 势能



$$G(q) = -(m_1 J_{vc1}^T g + m_2 J_{vc2}^T g + \cdots + m_n J_{vcn}^T g)$$



## 2.5 例题1

如图，一RP平面机器人，连杆质量分别为 $m_1, m_2$ ，连杆质心距关节1的距离分别为 $l_1, d_2$ ，连杆其质心处的惯性张量分别为：

$${}^{c_1}I_1 = \begin{bmatrix} I_{xx1} & 0 & 0 \\ 0 & I_{yy1} & 0 \\ 0 & 0 & I_{zz1} \end{bmatrix}, {}^{c_2}I_2 = \begin{bmatrix} I_{xx2} & 0 & 0 \\ 0 & I_{yy2} & 0 \\ 0 & 0 & I_{zz2} \end{bmatrix}$$

求此操作臂的动力学方程。

$$\blacksquare {}^0P_{c1} = \begin{bmatrix} -l_1 s_1 \\ l_1 c_1 \\ 0 \end{bmatrix}, {}^0P_{c2} = \begin{bmatrix} -(q_2 + d_2)s_1 \\ (q_2 + d_2)c_1 \\ 0 \end{bmatrix} \Rightarrow {}^0J_{vc1} = \begin{bmatrix} -l_1 c_1 & 0 \\ -l_1 s_1 & 0 \\ 0 & 0 \end{bmatrix}, {}^0J_{vc2} = \begin{bmatrix} -(q_2 + d_2)c_1 & -s_1 \\ -(q_2 + d_2)s_1 & c_1 \\ 0 & 0 \end{bmatrix}$$

$$\blacksquare {}^0J_{\omega c1} = [\bar{\epsilon}_1 Z_1 \quad 0] = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}, {}^0J_{\omega c2} = [\bar{\epsilon}_1 Z_1 \quad \bar{\epsilon}_2 Z_2] = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}$$

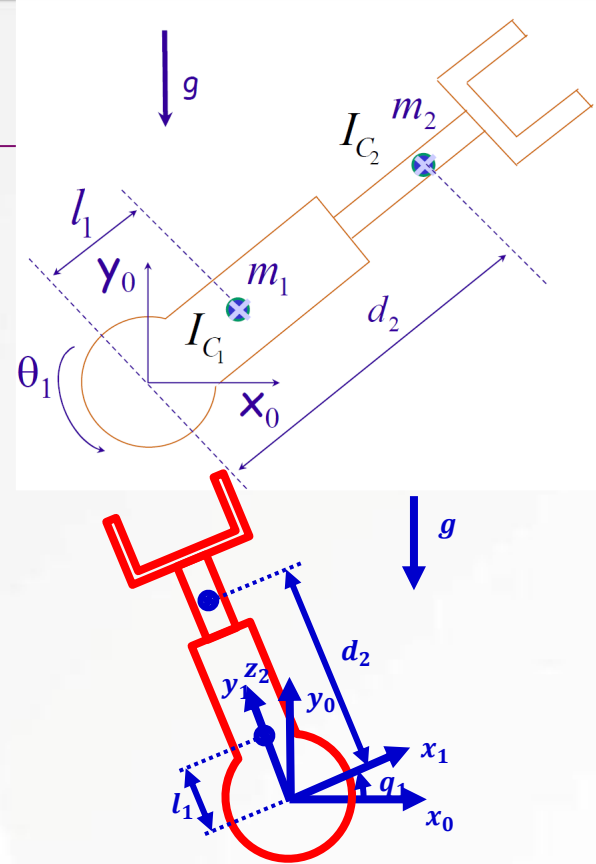
$$M(q) = \sum_{i=1}^n J_{vci}^T m_i J_{vci} + J_{\omega ci}^T I_{ci} J_{\omega ci} = \begin{bmatrix} m_1 l_1^2 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} I_{zz1} & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} m_2 (d_2 + q_2)^2 & 0 \\ 0 & m_2 \end{bmatrix} + \begin{bmatrix} I_{zz2} & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} I_{zz2} + I_{zz1} + m_1 l_1^2 + m_2 (d_2 + q_2)^2 & 0 \\ 0 & m_2 \end{bmatrix}$$

$$\blacksquare b_{i,j,k} = \frac{\partial m_{i,j}}{\partial q_k}, c_{ijk} = \frac{1}{2} (b_{i,j,k} + b_{i,k,j} - b_{j,k,i}):$$

$$b_{1,1,1} = 0, b_{1,1,2} = 2m_2(q_2 + d_2), b_{1,2,1} = b_{1,2,2} = b_{2,1,1} = b_{2,1,2} = b_{2,2,1} = b_{2,2,2} = 0$$

$$c_{111} = c_{122} = c_{222} = c_{212} = 0, c_{211} = -m_2(q_2 + d_2), c_{112} = m_2(q_2 + d_2)$$

$$V(q, \dot{q}) = \begin{bmatrix} c_{111} & c_{122} \\ c_{211} & c_{222} \end{bmatrix} \begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \end{bmatrix} + \begin{bmatrix} 2c_{112} \\ 2c_{212} \end{bmatrix} \dot{q}_1 \dot{q}_2 = \begin{bmatrix} 0 & 0 \\ -m_2(q_2 + d_2) & 0 \end{bmatrix} \begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \end{bmatrix} + \begin{bmatrix} 2m_2(q_2 + d_2) \\ 0 \end{bmatrix} \dot{q}_1 \dot{q}_2$$





## 2.5 例题1

$$\blacksquare {}^0P_{c1} = \begin{bmatrix} -l_1 s1 \\ l_1 c1 \\ 0 \end{bmatrix}, {}^0P_{c2} = \begin{bmatrix} -(q_2 + d_2)s1 \\ (q_2 + d_2)c1 \\ 0 \end{bmatrix} \Rightarrow {}^0J_{vc1} = \begin{bmatrix} -l_1 c1 & 0 \\ -l_1 s1 & 0 \\ 0 & 0 \end{bmatrix}, {}^0J_{vc2} = \begin{bmatrix} -(q_2 + d_2)c1 & -s1 \\ -(q_2 + d_2)s1 & c1 \\ 0 & 0 \end{bmatrix}$$

$$M(q) = \begin{bmatrix} I_{zz1} + I_{zz2} + m_1 l_1^2 + m_2 (d_2 + q_2)^2 & 0 \\ 0 & m_2 \end{bmatrix}$$

$$V(q, \dot{q}) = \begin{bmatrix} 0 & 0 \\ -m_2 (q_2 + d_2) & 0 \end{bmatrix} \begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \end{bmatrix} + \begin{bmatrix} 2m_2 (q_2 + d_2) \\ 0 \end{bmatrix} \dot{q}_1 \dot{q}_2$$

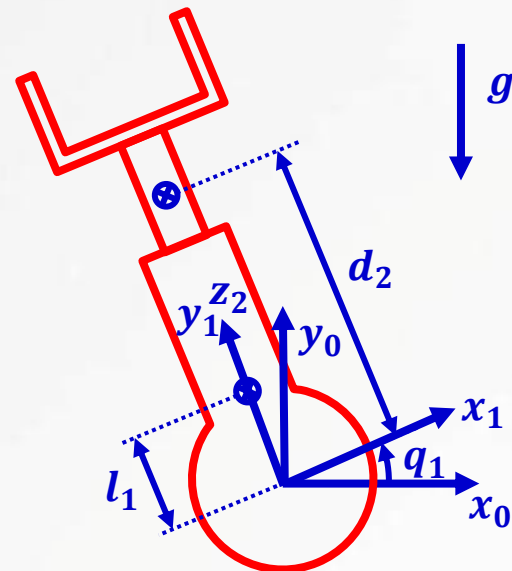
$$\blacksquare \text{重力项: } g = [0 \quad -g \quad 0]^T$$

$$\begin{aligned} G(q) &= -(m_1 J_{vc1}^T g + m_2 J_{vc2}^T g) = - \begin{bmatrix} -l_1 c1 & -l_1 s1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -m_1 g \\ 0 \end{bmatrix} - \begin{bmatrix} -(q_2 + d_2)c1 & -(q_2 + d_2)s1 & 0 \\ -s1 & c1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -m_2 g \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} -(m_1 l_1 + m_2 (q_2 + d_2)) g s1 \\ m_2 g c1 \end{bmatrix} \end{aligned}$$

■ 机器人动力学方程:

$$\begin{bmatrix} I_{zz2} + I_{zz1} + m_1 l_1^2 + m_2 (d_2 + q_2)^2 & 0 \\ 0 & m_2 \end{bmatrix} \ddot{q} + \begin{bmatrix} 0 & 0 \\ -m_2 (q_2 + d_2) & 0 \end{bmatrix} \begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \end{bmatrix} + \begin{bmatrix} 2m_2 (q_2 + d_2) \\ 0 \end{bmatrix} \dot{q}_1 \dot{q}_2 + \begin{bmatrix} -(m_1 l_1 + m_2 (q_2 + d_2)) g s1 \\ m_2 g c1 \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix}$$

$$\begin{bmatrix} I_{zz2} + I_{zz1} + m_1 l_1^2 + m_2 (d_2 + q_2)^2 & 0 \\ 0 & m_2 \end{bmatrix} \ddot{q} + \begin{bmatrix} 2m_2 (d_2 + q_2) \dot{q}_1 \dot{q}_2 \\ -m_2 (d_2 + q_2) \dot{q}_1^2 \end{bmatrix} + \begin{bmatrix} -(m_1 l_1 + m_2 (d_2 + q_2)) s1 \\ m_2 c1 \end{bmatrix} g = \Gamma$$



## 2.5 PUMA机器人动力学方程

$$\begin{aligned}
 D_{11} = & m_1 k_{1yy}^2 + \\
 & + m_2 (k_{2xx}^2 s_2^2 + k_{2yy}^2 c_2^2 + a_2^2 c_2^2 + 2a_2 c_2^2 x_2) \\
 & + m_3 [k_{3xx}^2 s_2^2 + k_{3xz}^2 c_2^2 + d_3^2 + a_2^2 c_2^2 + \\
 & + a_3^2 c_2^2 + 2a_2 a_3 c_2 c_{23} + 2x_3 (a_2 c_2 c_{23} + \\
 & + a_3 c_2^2) + 2y_3 d_3 + 2z_3 (a_3 c_2 s_{23} + \\
 & + a_2 c_2 s_{23})] + \\
 & + m_4 \{k_{4xx}^2 c_2^2 (s_2^2 + s_3^2 + 2s_2 s_3 c_{23}) + \\
 & + k_{4yy}^2 (c_2^2 - 2s_2 s_3 c_{23} - s_3^2) + k_{4xz}^2 s_4^2 (s_2^2 + \\
 & + s_3^2 + 2s_2 s_3 c_{23}) - 2y_4 s_{23} (c_2 a_2 + s_{23} d_4 + \\
 & + c_{23} a_3) - 2z_4 [a_3 s_4 (s_2 s_{23} - c_2^2) - \\
 & - c_2 s_4 c_{23} a_2 - s_4 c_{23} s_{23} d_4 + c_4 d_3] + d_3^2 + \\
 & + c_2^2 a_2^2 + s_{23}^2 d_4^2 + (1 - s_{23}^2) a_3^2 + 2c_2 c_{23} a_2 a_3 \\
 & + 2c_2 s_{23} a_2 d_4 + 2c_{23} s_{23} a_3 d_4\} + \\
 & + m_5 \{k_{5xx}^2 [s_{23}^2 (1 - s_4^2 c_5^2) - (s_2^2 - c_2^2) (s_5^2 + \\
 & + 2c_3 s_3 c_4 c_5 s_5) - 2(s_3^2 - c_3^2) c_2 s_2 c_4 c_5 s_5 - \\
 & - 2s_5^2 (s_3^2 + 2s_2 s_3 c_{23})] + k_{5yy}^2 s_4^2 (s_2 c_3 s_{23} + \\
 & + s_3^2 c_2^2) + k_{5xz}^2 [c_{23}^2 + (s_2^2 - c_2^2) (s_5^2 + \\
 & + 2c_3 s_3 c_4 c_5 s_5) + 2(s_3^2 - c_3^2) c_2 s_2 c_4 c_5 s_5 + \\
 & + s_5^2 (2s_3^2 + 4s_2 s_3 c_{23} - s_4^2 s_{23}^2)] +
 \end{aligned}$$

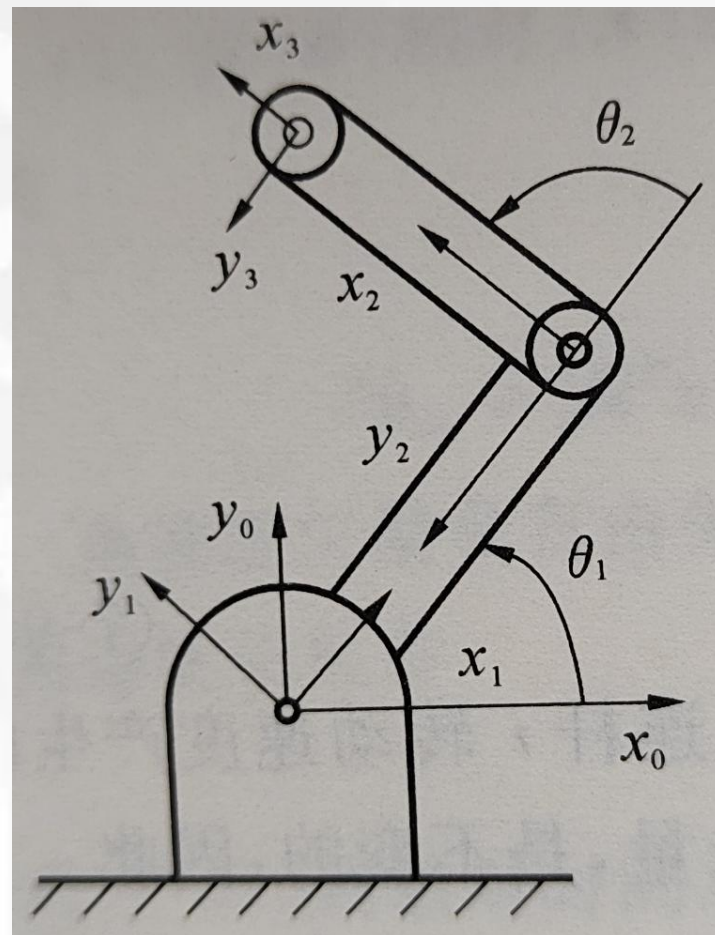
$$\begin{aligned}
 & + 2[(s_2 s_{23} - c_3^2) a_2 - c_{23}^2 a_3 - (c_2 s_2 + c_3 s_3 - \\
 & - 2s_2 s_3 s_{23}) d_4] s_4 + c_4 d_3] y_5 + 2[(s_2 c_{23} s_5 - \\
 & + s_2 s_5 s_{23} (1 - c_4) + c_3 c_4 s_5 + s_3 c_5) a_2 + \\
 & + (c_4 s_5 c_{23}^2 + (c_2 s_2 + c_3 s_3 - \\
 & - 2s_2 s_3 s_{23}) c_5) a_3 + (c_4 s_5 (c_2 s_2 + c_3 s_3 - \\
 & - 2s_2 s_3 s_{23}) + c_5 (s_2^2 + s_3^2 + 2s_2 s_3 c_{23})) d_4 + \\
 & + s_4 s_5 d_3] z_5 + d_3^2 + c_2^2 a_2^2 + c_{23}^2 a_3^2 + \\
 & + (1 - c_{23}^2) d_4^2 + 2c_2 s_{23} a_2 d_4 + \\
 & + 2(c_3 - s_2 s_{23}) a_2 a_3 + \\
 & + 2(c_2 s_2 + c_3 s_3 - 2s_2 s_3 s_{23}) a_3 d_4\} + \\
 & + m_6 \{k_{6xx}^2 [(c_{23}^2 - 1) (s_4^2 (1 - 2s_6^2) + c_6^2 \cdot \\
 & (s_5^2 c_4^2 - c_5^2) + 2c_4 s_4 c_5 c_6 s_6) + s_5 c_6 (s_5 c_6 + \\
 & + 2(c_4 c_5 c_6 - s_4 s_6) (c_2 s_2 + c_3 s_3 - \\
 & - 2s_2 s_3 s_{23}))] + k_{6yy}^2 [(c_{23}^2 - 1) (s_4^2 (2s_6^2 - \\
 & - 1) - s_6^2 (c_5^2 - c_5^2 s_5^2) - 2c_4 s_4 c_5 c_6 s_6) + \\
 & + s_5 s_6 (s_5 s_6 + 2(s_4 c_6 + c_4 c_5 s_6) (c_2 s_2 + \\
 & + c_3 s_3 - 2s_2 s_3 s_{23}))] + k_{6xz}^2 [c_{23}^2 c_5^2 - \\
 & - c_4^2 s_5^2 (c_{23}^2 - 1) - 2c_4 c_5 s_5 (c_2 s_2 + c_3 s_3 -
 \end{aligned}$$

$$\begin{aligned}
 & - 2s_2 s_3 s_{23})] + 2z_6 [(c_4 (c_3 s_5 - s_2 c_6 s_{23}) + \\
 & + c_5 (s_3 + s_2 c_{23})) a_2 + (c_4 s_5 c_{23}^2 + c_5 (c_2 s_2 + \\
 & + c_3 s_3 - 2s_2 s_3 s_{23})) a_3 + s_4 s_5 d_3 + (c_4 s_5 (c_2 s_2 + \\
 & + c_3 s_3 - 2s_2 s_3 s_{23}) - c_5 (c_{23}^2 - 1)) d_4] + \\
 & + c_2^2 a_2^2 + c_{23}^2 a_3^2 + d_3^2 - (c_{23}^2 - 1) d_4^2 + 2(c_3 - \\
 & - s_2 s_{23}) a_2 a_3 + 2(s_2 c_{23} + s_3) a_2 d_4 + \\
 & + 2(c_2 s_2 + c_3 s_3 - 2s_2 s_3 s_{23}) a_3 d_4\}
 \end{aligned}$$

共计60( $n \times 10$ )个相关参数

# 作业

如下图所示平面2R机械臂，分别用牛顿-欧拉法和拉格朗日法推导其动力学方程，写出详细推导过程





南开大学  
Nankai University



天津市智能机器人技术重点实验室  
Tianjin Key Laboratory of Intelligent Robotics  
南开大学机器人与信息自动化研究所  
Institute of Robotics & Automatic Information System

# 第六章 机器人动力学

## 《机器人学导论》

袁明星 副教授

Email: [mxyuan@nankai.edu.cn](mailto:mxyuan@nankai.edu.cn)

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